

Modules

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10.1 Definitions and examples

Def let R be a ring. A left R -module over R is a set M with:

- (1) $(+, M)$ M abelian
- (2) $R \ni r, m \in M \mapsto rm \in M$ satisfies:
 - (a) $(r+s)m = rm + sm \quad \forall r, s \in R, m \in M$.
 - (b) $(rs)m = r(sm) \quad \forall r, s \in R, m \in M$
 - (c) $r(m+n) = rm + rn \quad \forall r \in R, m, n \in M$.

If the ring has a 1 then:

$$(d) 1m = m \quad \forall m \in M$$

Modules over a field F and vector spaces over F are the same.

Def let R be a ring. let M be an R -module.
An R -submodule is a subgroup N of M which is closed under R action
i.e. $rn \in N \quad \forall r \in R, n \in N$

Examples

- (1) let R be any ring. $M=R$ is a left R -module.
Submodules are the ideals of R . (left) (right)
In particular, even F is a one dim vector space over itself
If R is non commutative then it may have different left, right structures.

- (2) let $R=F$ a field.
let $n \in \mathbb{Z}^+$ and let

$$F^n = \{(a_1, \dots, a_n) \mid a_i \in F \forall i\} \quad (\text{affine } n\text{-space})$$

$$\text{def } + \text{ on } F^n \quad (a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n)$$

$$d(a_1, \dots, a_n) = (da_1, \dots, da_n) \quad d \in F$$

Affine n -space is a vector space of dim n over F .

- (3) let R be a ring with 1. let $n \in \mathbb{Z}^+$.

$$R^n = \{(a_1, \dots, a_n) \mid a_i \in R, \forall i\}.$$

$$\text{def } + : (a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n)$$

$$\times : d(a_1, \dots, a_n) = (da_1, \dots, da_n) \quad d \in R$$

check axioms:

(*) $(+, R^n)$ is abelian

(a) let $d, \beta \in R$ let $(a_1, \dots, a_n) \in R^n$

$$(d+\beta)(a_1, \dots, a_n) = ((d+\beta)a_1, \dots, (d+\beta)a_n) = (da_1 + \beta a_1, \dots, da_n + \beta a_n) = d(a_1, \dots, a_n) + \beta(a_1, \dots, a_n)$$

$$(b) (d\beta)(a_1, \dots, a_n) = (d\beta a_1, \dots, d\beta a_n)$$

$$= (d(\beta a_1), \dots, d(\beta a_n))$$

$$= d(\beta(a_1, \dots, a_n))$$

$$(c) d((a_1, \dots, a_n) + (b_1, \dots, b_n)) = d(a_1 + b_1, \dots, a_n + b_n)$$

$$= (d(a_1 + b_1), \dots, d(a_n + b_n))$$

$$= (da_1 + db_1, \dots, da_n + db_n)$$

$$= d(a_1, \dots, a_n) + d(b_1, \dots, b_n)$$

$$(d) 1(a_1, \dots, a_n) = (a_1, \dots, a_n)$$

$\Rightarrow$ check complete.

R^n is called free module of rank n over R .

$\{(0, \dots, a_i, \dots, 0)\}$ is a submodule $a_i \in R$ of R^n .

- (4) if M is an R -module and S is a subring of R with $1_S = 1_R$ then M is an S -module

$S \ni s, m \in M \mapsto sm \in M$ satisfies all the axioms.

$$1_S m = 1_R m = m.$$

field R is an R -module, $\mathbb{Z}$ -module, $\mathbb{Q}$ -module.

- (5) Suppose M is an R -module and I a 2-sided ideal of R with $I \subseteq \text{ann}(M)$

Then $\forall m \in M$ and $\bar{r} \in R/I$ let

$$\bar{r}m = (r+I)m = rm$$

Then M is an R/I module.

In particular if I is maximal in a commutative ring R and $IM=0$

$\Rightarrow (R/I)$ is a field and M is a vectorspace over R/I .

$\mathbb{Z}$ -modules

Let $R=\mathbb{Z}$. let $(A,+)$ be any abelian group.

$$\mathbb{Z} \ni a, + : \forall n \in \mathbb{Z}, a \in A \text{ define.}$$

$$na = \begin{cases} a + \dots + a & n > 0 \\ 0 & n = 0 \\ -a - \dots - a & n < 0 \end{cases}$$

(n) times

This is the only way $\mathbb{Z}$ acts on A making it unital $\mathbb{Z}$ -module.

Thus, every abelian group is a $\mathbb{Z}$ -module.

Conversely, if M is any $\mathbb{Z}$ module, M is an abelian group.

$$\{\mathbb{Z}\text{-modules}\} = \{\text{abelian groups}\}$$

$$\Rightarrow \{\mathbb{Z}\text{-submodules}\} = \{\text{subgroups}\}$$

• $\langle a \rangle$ is a $\mathbb{Z}$ -module.

$$(a) m_1 + m_2 \cdot a = a^{m_1 + m_2} = a^{m_1} \cdot a^{m_2}$$

$$(b) (m_1 m_2) \cdot a = a^{(m_1 m_2)} = (a^{m_1})^{m_2}$$

$$(c) m_2(a^{n_1} a^{n_2}) = (a^{n_1} a^{n_2})^m = (a^{n_1 + n_2})^m = a^{mn_1} a^{mn_2}$$

$$(d) 1 \cdot a = a^1 = a$$

If A abelian s.t. $\exists x \in A$ of finite order then $nx = 0$. Thus, $\mathbb{Z}$ -module may have non-zero elements such that $nx = 0$ for some nonzero ring element n . In particular, if A has order n , then by Lagrange's theorem $nx = 0 \forall x \in A$

$\Rightarrow A$ is a module over $\mathbb{Z}/n\mathbb{Z}$

If p is prime, then A is a vector space over $\mathbb{Z}/p\mathbb{Z}$.

Klein 4-group $\cong \mathbb{F}_2^2$ is a vector space over $\mathbb{F}_2$

Examples ($F[x]$ module.)

Let F be a field. Let x be indeterminate. Let $R = F[x]$.

Let V be a vector space over F . Let $T \in \text{End}(V)$

$$T^0 = I$$

$$\vdots$$

$$T^n = T \circ T \circ \dots \circ T \text{ (n times)}$$

$$\forall A, B \in \text{End}(V) \text{ and } \alpha, \beta \in F$$

$$(\alpha A + \beta B)(v) = \alpha(A(v)) + \beta(B(v)) \quad [\text{End}(V) \text{ is a vector space over } F]$$

$$\underline{\text{Def}} \quad F[x] \ni v : \forall v \in V$$

$$p(x)v = (a_n T^n + \dots + a_0)v$$

$$= a_n T^n(v) + \dots + a_0 v$$

$\Rightarrow V$ is an $F[x]$ module.

Action of $F[x]$ on V is dependent on the choice of $T \in \text{End}(V)$.

$\Rightarrow$ There are many $F[x]$ module structures.

If $T = 0$ then $p(x)v = a_0 v \quad \forall p(x) \in F[x]$

$\Rightarrow F[x]$ module structure is just F module structure.

If $T = I$.

$$p(x)v = (a_n + \dots + a_0)v = a_n v + \dots + a_0 v \quad \forall p(x) \in F[x].$$

• Example: let V be $\mathbb{F}^n$ let $T: \mathbb{F}^n \rightarrow \mathbb{F}^n$

$$T(x_1, \dots, x_n) = (x_2, x_3, \dots, 0)$$

• let $\{e_i\}$ be the canonical basis.

$$T^k(e_i) = \begin{cases} e_{i-k} & \text{if } i > k \\ 0 & \text{if } i \leq k \end{cases}$$

$$\text{if } m < n$$

$$(a_m x^m + \dots + a_0)e_n$$

$$= a_m T^m(e_n) + \dots + a_0 e_n$$

$$= a_m e_{n-m} + a_{m-1} e_{n-(m-1)} + \dots + a_0 e_n$$

$$= (0, 0, \dots, 0, a_m, a_{m-1}, \dots, a_0)$$

$\Rightarrow$ we can determine the action of whole of $F[x]$ on V .

$F[x]$ module is a vector space together with $T \in \text{End}(V)$

Actions of F and x on V uniquely determine $F[x] \ni v$.

$$\{V \text{ an } F[x] \text{ module}\} \longleftrightarrow \left\{ \begin{array}{l} V \text{ vector space over } F \\ \text{and} \\ T \in \text{End}(V) \end{array} \right\}$$

the element x acts on V as $T \in \text{End}(V)$

submodules of $F[x]$ module V .

If W an $F[x]$ submodule of $F[x]$ module V

Then, W is an F -submodule of V i.e. subspace of V

and W is invariant under action of x i.e. $T(W) \subseteq W$. (T -invariant)

$$\Rightarrow T^n(W) \subseteq W \quad \forall n \in \mathbb{Z}^+$$

$\Rightarrow p(x) \ni W$ is an invariant action.

$$p(T)W \subseteq W.$$

• $F[x]$ submodules of V are T invariant subspaces.

$$\{W \text{ an } F[x] \text{ submodule}\} \leftrightarrow \left\{ \begin{array}{l} W \text{ subspace of } V \\ W \text{ is invariant under } T \end{array} \right\}$$

if T is the shift operator:

$$U_k = \{(x_1, x_2, \dots, x_k, 0, 0, \dots) \mid x_i \in F\} \quad 0 \leq k \leq n \text{ is clearly } T\text{-invariant}$$

$$\Rightarrow F[x] \text{ submodule of } V.$$

Prop 1 Let R be a ring. Let M be an R -module.

$N \subseteq M$ is a submodule $\Leftrightarrow$

$$(1) N \neq \emptyset$$

$$(2) x + ry \in N \quad \forall r \in R, \forall x, y \in N.$$

Def $(\Rightarrow)$ If N is a submodule then $0 \in N$ (subgroups)
 $ry \in N \quad \forall r \in R, y \in N$
 $\Rightarrow x + ry \in N.$

$(\Leftarrow)$ Let $r=1 \Rightarrow x+y \in N \quad \forall x, y \in N \Rightarrow N$ is a subgroup of M .

$$\Rightarrow 0 \in N$$

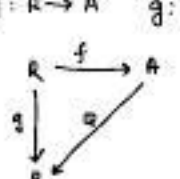
$$\Rightarrow ry \in N \quad \forall r \in R, y \in N. \Rightarrow N \subseteq M \text{ is a submodule.}$$

Def R commutative with 1. An R -algebra is a ring A with 1 together with ring hom $f: R \rightarrow A$ $f(1_R) = 1_A$ s.t. $f(R) \subseteq \text{Centre}(A)$

A has natural left and right module structures (unital) $f(r)a = r \cdot a = a \cdot r = af(r)$

Def A, B are R -algebras. R -algebra homomorphism is $\varphi: A \rightarrow B$ with $\varphi(1_A) = 1_B$ s.t. $\varphi(ra) = r \cdot \varphi(a) \quad \forall r \in R, \forall a \in A$

$$\left[\begin{array}{l} A, f \\ B, g \end{array} \right. \quad \varphi(ra) = \varphi(f(r)a) = \varphi(f(r)) \cdot \varphi(a) = g(r) \cdot \varphi(a)$$

$$\varphi \circ f = g \quad f: R \rightarrow A \quad g: R \rightarrow B$$


Examples Let R be a commutative ring with 1.

(1) any ring with 1 is a $\mathbb{Z}$ -algebra. (Abelian group structure of the ring)

(2) $\forall A$ ring with 1, if $R \subseteq C(A)$, then A is an R -algebra.
 $s.t. 1_A \in R$

In particular, if A is commutative ring with 1, then A is an R -algebra for any $R \subseteq A$ subring with 1.

$R[x]$ is an R -algebra. $R[x_1, \dots, x_n]$ is an R -algebra [since $R \subseteq R[x]$ $1 \in R$.]
 $R[G]$, G finite group is an R -algebra

(3) If A is an R -algebra, $f(R) \subseteq C(A)$ as subring containing 1_A .

A 's R -module structure is dependent only on the subring $f(R) \subseteq C(A)$ upto ring hom, every algebra A arises from subring of $C(A)$ that has 1_A .

(4) When $R=F$ is a field. $f(F) \cong F \subseteq A$ as subring.

A is an algebra over a field $F = A$ contains the field F in the centre.
 or $F \subseteq C(A)$ and $1_F = 1_A$

suppose A is an R -algebra. Then, A is a ring with identity which is a unital left R -module.
 satisfying $r(ab) = (r \cdot a)b = a(r \cdot b) \quad \forall r \in R$ and $a, b \in A$.

Exercises R with $1 \in R$. M left R -module.

$$(1) 0m = 0$$

$$\text{Def } 0m = (0+0)m = 0m + 0m \Rightarrow 0m = 0$$

$$\text{Def } (-1)m = -m$$

$$0 = (-1+1)m = (-1)m + 1m = -m + m \Rightarrow (-1)m = -m.$$

$$(2) R^* \supseteq M \quad R^* \subseteq R \text{ module axioms.}$$

(3) Suppose $rm = 0$ for some $r \in R$ and some $m \in M$. $m \neq 0$.

Suppose $\exists r^{-1} \in R$. Then,

$$m = r^{-1}rm = r^{-1}0 = 0 \quad \text{contradiction}$$

(4) $M = R^n$ (R ring with 1) $I_1, \dots, I_n$ be ideals of R .

$$(a) \{(x_1, \dots, x_n) \mid x_i \in I_i\} = U$$

$$0 \in I_i \quad \forall i \Rightarrow (0, \dots, 0) \in U \Rightarrow U \neq \emptyset$$

$$\bar{x} \in U, \bar{y} \in U, r \in R$$

$$r\bar{y} = r(y_1, \dots, y_n) = (ry_1, \dots, ry_n) \quad ry_i \in I_i \quad \text{since } I_i \text{ ideal of } R$$

$$\Rightarrow r\bar{y} \in U.$$

$$\bar{x} + r\bar{y} \in U. \Rightarrow U \text{ submodule.}$$

$$(b) \{(x_1, \dots, x_n) \mid x_i \in R \text{ and } x_1 + \dots + x_n = 0\} = U$$

$$0+0+\dots+0=0 \in U \Rightarrow U \neq \emptyset$$

$$(x_1, \dots, x_n) \in U, (y_1, \dots, y_n) \in U, w \in R.$$

$$(x_1, \dots, x_n) + r(y_1, \dots, y_n) = (x_1 + ry_1, \dots, x_n + ry_n)$$

$$x_1 + ry_1 + \dots + x_n + ry_n = (x_1 + \dots + x_n) + r(y_1 + \dots + y_n) = 0 + r \cdot 0 = 0$$

$\Rightarrow U$ is a submodule of M .

(5) $\forall I$ left ideal of R def:

$$I = \{ \sum a_i m_i \mid a_i \in R, m_i \in I \}$$

$$IM = \left\{ \sum_{i=1}^n a_i m_i \mid a_i \in I, m_i \in M \right\}$$

$$0 \in IM \quad (0m)$$

$$\sum_{i=1}^n a_i m_i + r \sum_{j=1}^n b_j m_j = \sum_{i=1}^n a_i m_i + \sum_{j=1}^n (r b_j) m_j = \sum_{k=1}^n a_k m_k \in IM. \Rightarrow IM \text{ submodule.}$$

(6) $\mathcal{A}$ be a collection of submodule of an R -module M .

$$\bigcap_{A \in \mathcal{A}} A \neq \emptyset \text{ since } 0 \in A \forall A \in \mathcal{A}$$

$$\text{let } x, y \in \bigcap_{A \in \mathcal{A}} A \text{ let } r \in R.$$

$$x + ry \in A \forall A \in \mathcal{A}.$$

$$\Rightarrow x + ry \in \bigcap_{A \in \mathcal{A}} A \Rightarrow \bigcap_{A \in \mathcal{A}} A \text{ is a submodule.}$$

(7) $N_1 \subseteq N_2 \subseteq \dots$ be an increasing chain of submodules of M .

$$N_1 \cup N_2 = N_2 \text{ is a submodule}$$

$$\text{Suppose } \bigcup_{i=1}^k N_i \text{ is a submodule.}$$

$$\bigcup_{i=1}^{k+1} N_i = N_{k+1} \cup \left(\bigcup_{i=1}^k N_i \right) = N_{k+1} \text{ is a submodule.}$$

$$\Rightarrow \bigcup_{i=1}^{\infty} N_i \text{ is a submodule.}$$

(8) $m \in M$ on R -module is called a torsion element if $rm = 0$ for some $r \neq 0$ in R .

$$\text{Tor}(M) = \{ m \in M \mid rm = 0 \text{ for some } r \neq 0 \in R \}$$

(a) Suppose R is an integral domain

$$0 \in \text{Tor}(M) \text{ since } 0 \in M \text{ and } r \cdot 0 = 0 \forall r \in R.$$

$$\text{let } m, n \in \text{Tor}(M) \exists r_1, r_2 \neq 0 \in R \text{ s.t. } r_1 m = 0, r_2 n = 0 \quad r_1 r_2 \neq 0 \text{ since } R \text{ integral domain}$$

$$\text{let } r \in R$$

$$r_1 r_2 (m + rn) = r_1 r_2 m + r_1 r_2 r n = r_2 (r_1 m) + r r_1 (r_2 n) = 0$$

$$\Rightarrow m + rn \in \text{Tor}(M) \Rightarrow \text{Tor}(M) \text{ is a submodule of } M.$$

$$(b) \text{Tor}(\mathbb{Z}/6\mathbb{Z}) = \{0, 2, 3, 4\}$$

$$2 + 3 = 5 \notin (\mathbb{Z}/6\mathbb{Z})^\times \Rightarrow \text{Tor}(\mathbb{Z}/6\mathbb{Z}) \text{ is not a submodule.}$$

(c) Suppose R has zero divisors.

$$\text{let } M \text{ be an } R\text{-module. } M \neq \{0\}$$

$$\text{Suppose } r_1 r_2 = 0, r_1, r_2 \neq 0$$

$$0 = (r_1 r_2)m = r_1(r_2 m) \forall m \in M$$

$$\text{if } r_2 m \neq 0 \neq 0 \text{ then } m \in \text{Tor}(M)$$

$$\text{if } r_2 m = 0 \Rightarrow r_2 m \in \text{Tor}(M)$$

$$\Rightarrow \text{Every nonzero module has nonzero torsion elements.}$$

(9) If N is a submodule M , $\text{ann}_R(N) := \{r \in R \mid rn = 0 \forall n \in N\}$

$$0 \in \text{ann}_R(N)$$

$$\text{let } r_1, r_2 \in \text{ann}_R(N) \text{ let } n \in N$$

$$(r_1 - r_2)n = r_1 n - r_2 n = 0$$

$$\text{let } r \in R \text{ let } r_1 \in \text{ann}_R(N)$$

$$(r r_1)n = r(r_1 n) = r \cdot 0 = 0 \Rightarrow r r_1 \in \text{ann}_R(N)$$

$$(r_1 r)n = r_1(rn) = 0 \text{ since } rn \in N. \Rightarrow r_1 r \in \text{ann}_R(N)$$

$$\Rightarrow \text{ann}_R(N) \text{ is a 2-sided ideal.}$$

(10) let I be a right ideal of R ,

$$\text{ann}_M(I) := \{m \in M \mid am = 0 \forall a \in I\}$$

$$0 \in \text{ann}_M(I)$$

$$m, n \in \text{ann}_M(I) \text{ let } r \in R, \text{ let } a \in I$$

$$a(m + rn) = am + arn = 0 \quad ar \in I \text{ (I right ideal)}$$

$$\Rightarrow m + rn \in \text{ann}_M(I)$$

$$\Rightarrow \text{ann}_M(I) \text{ is a submodule of } M.$$

$$(11) M = \mathbb{Z}/24\mathbb{Z} \times \mathbb{Z}/15\mathbb{Z} \times \mathbb{Z}/50\mathbb{Z} \text{ as } \mathbb{Z}\text{-module.}$$

$$(a) \text{ann}_{\mathbb{Z}}(M) \text{ is an ideal in } \mathbb{Z} \Rightarrow \text{ann}_{\mathbb{Z}}(M) = (a) \quad a \in \mathbb{Z} \text{ since } \mathbb{Z} \neq 0.$$

$$a = \text{lcm}(24, 15, 50)$$

$$(b) \text{let } I = 2\mathbb{Z} \quad 2x \equiv 0 \pmod{24}, 2y \equiv 0 \pmod{15}, 2z \equiv 0 \pmod{50}.$$

$$\begin{aligned} \text{ann}_M(2\mathbb{Z}) &= \left\langle \frac{24}{\gcd(24, 2)} \right\rangle \times \left\langle \frac{15}{\gcd(15, 2)} \right\rangle \times \left\langle \frac{50}{\gcd(50, 2)} \right\rangle \\ &= \langle 12 \rangle \times \langle 0 \rangle \times \langle 25 \rangle \end{aligned}$$

(12) (a) N submodule of M let $I = \text{ann}_R(N)$

$$\text{claim: } N \subseteq \text{ann}_M(I) = \{m \in M \mid am = 0 \forall a \in I\}$$

$$\text{Proof: } \forall n \in N \quad an = 0 \forall a \in I. \Rightarrow N \subseteq \text{ann}_M(I).$$

$$M = \mathbb{Z}/6\mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z}$$

$$R = \mathbb{Z}/6\mathbb{Z}$$

$$N = \langle (2, 0) \rangle = \langle 2 \rangle \times \{0\}$$

$$\text{ann}_R(N) = (3)$$

$$\text{ann}_M((1)) = \langle (2, 0), (0, 2) \rangle \neq \langle (2, 0) \rangle$$

(b) I right ideal of R $N = \text{ann}_M(I)$

$$\text{ann}_R(N) = \{r \in R \mid rn = 0\}$$

$$rN = 0 \Rightarrow I \subseteq \text{ann}_R(N)$$

$$R = \mathbb{Z} \quad M = \mathbb{Z}/8\mathbb{Z}$$

$$I = (6) \quad N = \text{Ann}_M(I) = (4)$$

$$\text{Ann}_R(N) = (2) \supset (6)$$

$$(13). \quad I \text{ is an ideal of } R. \quad M' = \{a \in M \mid \exists k \text{ s.t. } I^k a = 0\}$$

$$M' = \bigcup_{k=1}^{\infty} \text{Ann}_M(I^k)$$

$$\text{let } a \in \text{Ann}_M(I^k) \Rightarrow I^k a = 0$$

$$\Rightarrow I^{k+1}(a) = I(I^k a) = 0$$

$$\Rightarrow a \in \text{Ann}_M(I^{k+1})$$

Thus, $\text{Ann}_M(I) \subseteq \text{Ann}_M(I^2) \subseteq \dots$ is an ascending chain of submodules $\Rightarrow M'$ is a submodule.

$$(14) \quad \text{let } z \in C(R)$$

$$0 \in \mathbb{Z}M \quad (z0)$$

$$\text{let } x, y \in \mathbb{Z}M, r \in R.$$

$$x + ry = zx' + rzy' = z(x' + ry') \in \mathbb{Z}M. \quad (x', y' \in M)$$

$$\Rightarrow \mathbb{Z}M \text{ is a submodule.} \quad (x' + ry' \in M)$$

$$\left[\begin{array}{l} z \in C(R) \\ \Rightarrow \mathbb{Z}M \text{ is a submodule} \end{array} \right]$$

$$R = M_2(F) \quad e = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$eR = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \mid a \in F \right\}$$

$$\text{suppose } \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} \in eR. \quad \text{let } \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \in R.$$

$$\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a'b + a0 & \\ c'b & 0 \end{bmatrix} \notin eR.$$

$$\Rightarrow eR \text{ is not a submodule.}$$

$$\Rightarrow e \notin C(M_2(F))$$

$$(15). \quad \text{NO!} \quad \text{If } M \text{ is } \mathbb{Q}\text{-module then } M \text{ is vector space over } \mathbb{Q}$$

$$\text{suppose } m \in M \text{ with order } k.$$

$$km = 0$$

$$1 = \frac{k}{k} = \frac{1}{k}(km) = 0 \quad \downarrow$$

$$(16). \quad \text{Suppose } W \text{ is an } R[x]\text{-submodule} \Rightarrow W \text{ is a vector subspace of } V \text{ and } W \text{ is } T\text{-invariant. (T shift operator)} \quad (*)$$

$$\text{let } W \neq 0. \text{ be an invariant subspace of } V \text{ under } T$$

$$W = \text{sp}(v_1, \dots, v_k) \quad k \in \mathbb{N}$$

$$v_i = \sum_{j=1}^n a_{ij} e_j$$

$$k = \max \{i \mid a_{ij} \neq 0\}$$

$$\text{Then, each } v_i \in U_k \Rightarrow W \subseteq U_k$$

$$\text{let } w \in W \text{ s.t. } a_{ik} \neq 0$$

$$w = a_1 e_1 + \dots + a_k e_k \quad a_k \neq 0$$

$$T(w) = a_k e_1 \Rightarrow e_1 \in W$$

$$a_2 e_2 + \dots + a_k e_k \in W$$

$$\text{Then applying operator repeatedly like this, } [T^{k-2}(a_2 e_2 + \dots + a_k e_k) = a_{k-1} e_1 + a_k e_2 \Rightarrow e_2 \in W]$$

$$e_1, e_2, \dots, e_k \in W$$

$$\Rightarrow U_k \subseteq W.$$

$$(17). \quad \text{let } T \text{ be shift operator. } \{e_1, \dots, e_n\} \text{ be the usual basis of } V.$$

$$\text{if } m \geq n$$

$$(a_m T^m + a_{m-1} T^{m-1} + \dots + a_0) e_n = a_m T^m(e_n) + a_{m-1} T^{m-1}(e_n) + \dots + a_n T^n(e_n) + \dots + a_0 e_n$$

$$= 0 + 0 + \dots + 0 + a_{n-1} e_1 + a_{n-2} e_2 + \dots + a_0 e_n$$

$$= (a_{n-1}, a_{n-2}, \dots, a_0)$$

$$(18). \quad \text{let } F = \mathbb{R} \quad V = \mathbb{R}^2$$

$$T: V \rightarrow V$$

$$(x, y) \mapsto (y, -x)$$

$$W \text{ is an } R[x] \text{ submodule} \Rightarrow W \text{ is a subspace of } V \text{ and } W \text{ is invariant under } T.$$

$$\text{suppose } W \neq 0, W \neq V$$

$$\text{suppose } W = \text{sp}(v) \quad v = a_1 e_1 + a_2 e_2 = (a_1, a_2) \text{ invariant under } T. \quad a_1, a_2 \text{ not both zero.}$$

$$T(a_1, a_2) = (a_2, -a_1) = a_2 e_1 - a_1 e_2 \in \text{sp}(v)$$

$$\Rightarrow \exists \alpha \in \mathbb{R} \text{ s.t. } (a_2, -a_1) = (\alpha a_1, \alpha a_2)$$

$$\Rightarrow \alpha a_1 = a_2, \quad \alpha a_2 = -a_1$$

$$\Rightarrow \alpha = \frac{a_2}{a_1} \Rightarrow a_2^2 = -a_1^2 \Rightarrow a_2 = a_1 = 0 \quad \downarrow$$

$$\Rightarrow W = 0 \text{ or } W = V \text{ are the only submodules for } T.$$

$$(19). \quad F = \mathbb{R} \quad V = \mathbb{R}^2$$

$$T: V \rightarrow V$$

$$(x, y) \mapsto (0, y)$$

$$W = \text{sp}(v_0) \quad v_0 = (a_1, a_2)$$

$$T(v) = (0, a_2)$$

$$(0, a_2) \in W \Leftrightarrow a_1 = 0 \text{ or } a_2 = 0$$

$$\left[\begin{array}{l} (0, a_2) = (\alpha a_1, \alpha a_2) \quad \alpha \neq 0 \Rightarrow a_1 = 0 \\ \text{or } \alpha = 0 \Rightarrow a_2 = 0 \end{array} \right]$$

$\Rightarrow$ x-axis, y-axis are the only nontrivial submodules.

(20).
$$\begin{bmatrix} \omega_1(n) & 0 \\ 0 & \omega_2(n) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad T: (x, y) \mapsto (x, y)$$

 $\Rightarrow$ every subspace is invariant under T . $\Rightarrow$ every subspace is a submodule.

(21). $R = M_n(F) \quad n \geq 1$

let $M := \left\{ A \in M_n(F) \mid A_{ij} \in F \quad A_{ij} = 0 \quad n \geq j > i \right\}$

$$\left\{ \begin{bmatrix} A_{11} & 0 & 0 & \dots \\ 0 & \dots & \dots & \dots \\ \vdots & \vdots & \vdots & \vdots \\ A_{nn} & 0 & \dots & \dots \end{bmatrix} \right\}$$

$0 \in M$.

let $A, B \in M, \quad d \in R$

$(dA)_{ij} = \sum_{k=1}^n d_{ik} A_{kj}$

$B_{kj} = 0 \quad \forall n \geq k > j$

$\Rightarrow (dA)_{ij} = 0 \quad \forall n \geq j > i. \Rightarrow dA \in M$

$\Rightarrow A + dA \in M \quad ((A + dA)_{ij} = (A)_{ij} + (dA)_{ij} \Rightarrow (A + dA)_{ij} = 0 \quad \forall n \geq j > i$

$\Rightarrow M$ is a R -submodule, where R is a left module over itself.

however, $(BA)_{ij} = \sum_{k=1}^n B_{ik} A_{kj}$

$= B_{ii} A_{ij}$

$\Rightarrow (BA)_{ij} \neq 0 \quad \text{for } n \geq j > i \Rightarrow BA \notin M \Rightarrow M$ is not a submodule where R is a right submodule over itself.

(22). A ring with 1_R that is a unital left R -module

satisfying $r(ab) = (r \cdot a)b = a(r \cdot b)$

$f: R \rightarrow A \quad r \mapsto r \cdot 1_A$

$f(r_1 + r_2) = (r_1 + r_2) \cdot 1_A = r_1 \cdot 1_A + r_2 \cdot 1_A = f(r_1) + f(r_2)$

$f(r_1 r_2) = r_1 r_2 \cdot 1_A = r_1 r_2 \cdot (1_A \cdot 1_A) = r_1 \cdot 1_A r_2 \cdot 1_A = f(r_1) f(r_2)$

$\Rightarrow$ Ring hom.

$f(1_R) = 1_R \cdot 1_A = 1_A$

let $a \in f(R) \quad a = f(r) \quad \text{for some } r \in R$

let $b \in A \quad ab = f(r)(b \cdot 1_A) = r \cdot (b \cdot 1_A) = b \cdot (r \cdot 1_A) = b \cdot f(r) = ba \Rightarrow a \in CCR$

$\Rightarrow A$ is an R -algebra.

$\left[\begin{array}{l} a \in \ker(f) \\ f(a) = a \cdot 1_A = 0 \Rightarrow a = 0 \\ \Rightarrow \ker(f) = \{0\} \\ \Rightarrow f(R) \subseteq CCR \end{array} \right] \Rightarrow R\text{-module structure on } A = R\text{-module structure induced by its algebra structure.}$

(23). $A = \mathbb{C} \times \mathbb{C}$

$\tau_1 = \text{id}: \mathbb{C} \rightarrow \mathbb{C}$

$\tau_2 = *: \mathbb{C} \rightarrow \mathbb{C}$

$\forall (p, q) \in \{1, 2\} \times \{1, 2\}$

def $f_{p,q}: \mathbb{C} \rightarrow \mathbb{C} \times \mathbb{C} \quad f_{p,q}(z) = (\tau_p(z), \tau_q(z))$

(a) $f_{p,q}(z_1 + z_2) = (\tau_p(z_1 + z_2), \tau_q(z_1 + z_2)) = (\tau_p(z_1), \tau_q(z_1)) + (\tau_p(z_2), \tau_q(z_2))$
 $= f_{p,q}(z_1) + f_{p,q}(z_2)$

$f_{p,q}(z_1 z_2) = (\tau_p(z_1 z_2), \tau_q(z_1 z_2))$
 $= (\tau_p(z_1) \tau_p(z_2), \tau_q(z_1) \tau_q(z_2))$
 $= (\tau_p(z_1), \tau_q(z_1)) \cdot (\tau_p(z_2), \tau_q(z_2))$
 $= f_{p,q}(z_1) \cdot f_{p,q}(z_2)$

since τ_1, τ_2 are injective $\Rightarrow f_{p,q}$ is injective.

if $z \in \mathbb{R} \Rightarrow \tau_1(z) = \tau_2(z)$ since $*|_{\mathbb{R}} = \text{id}|_{\mathbb{R}}$

$\Rightarrow f_{p,q}$ all agree on $\mathbb{R} \Rightarrow f_{p,q}(1) = (1, 1)$

$\mathbb{C} \cong f_{p,q}(\mathbb{C}) \subseteq A \Rightarrow A$ has four distinct $\mathbb{C}$ -algebra structures

$z \cdot (u, v) = \begin{cases} f_{1,1}(z)(u, v) = (zu, zv) \\ f_{1,2}(z)(u, v) = (zu, \bar{z}v) \\ f_{2,1}(z)(u, v) = (\bar{z}u, zv) \\ f_{2,2}(z)(u, v) = (\bar{z}u, \bar{z}v) \end{cases}$

(b) Suppose $f_{p,q} \neq f_{p',q'}$

$\text{id}: A\text{-}(f_{p,q}\text{-algebra}) \rightarrow A\text{-}(f_{p',q'}\text{-algebra})$
 $(u, v) \mapsto (u, v)$

$\text{id}(z \cdot (u, v)) = \text{id}(f_{p,q}(z)(u, v)) = (\tau_p(z)u, \tau_q(z)v) \neq (f_{p',q'}(z)(u, v)) = (\tau_{p'}(z)u, \tau_{q'}(z)v)$
 $\neq \text{id}(f_{p',q'}(z)(u, v)) = f_{p',q'}(z)(u, v) = (\tau_{p'}(z)u, \tau_{q'}(z)v)$

$$\left[\begin{array}{l} \text{id}(i_{1,2}(1, 1)) = (i_{1,2}, i_{1,2}) \\ i_{2,1}(1, 1) = (i_{2,1}, i_{2,1}) \end{array} \right]$$

$\Rightarrow \text{id}$ is not a $\mathbb{C}$ -algebra hom from $A_{f_{p,q}}$ to $A_{f_{p',q'}}$

(c) $\phi: A_{f_{p,q}} \rightarrow A_{f_{1,1}}$

$(u, v) \mapsto (\tau_p(u), \tau_q(v))$

$\phi((u_1, v_1) + (u_2, v_2)) = (\tau_p(u_1 + u_2), \tau_q(v_1 + v_2))$
 $= (\tau_p(u_1), \tau_q(v_1)) + (\tau_p(u_2), \tau_q(v_2)) = \phi(u_1, v_1) + \phi(u_2, v_2)$

$(\tau_p(u_1 u_2), \tau_q(v_1 v_2)) = (\tau_p(u_1), \tau_q(v_1)) \cdot (\tau_p(u_2), \tau_q(v_2))$

$\Rightarrow$ ring hom.

since τ_p, τ_q are injective and surjective

$\Rightarrow \phi$ is a ring isomorphism.

let $z \in \mathbb{C}$

$\phi(z \cdot (u, v)) = \phi((f_{p,q}(z)(u, v))) = \phi((\tau_p(z)u, \tau_q(z)v))$

$$\begin{aligned}
&= \phi(\tau_p(z)u, \tau_q(z)v) \\
&= (\tau_p(\tau_p(z)u), \tau_q(\tau_q(z)v)) \\
&= (\tau_p(\tau_p(z))\tau_p(u), \tau_q(\tau_q(z))\tau_q(v)) \\
&= (\tau_p(\tau_p(z)), \tau_q(\tau_q(z))) \phi(u, v) \\
&= (z, z) \phi(u, v) \\
&= z_{1,2} \cdot \phi(u, v) \Rightarrow \mathbb{C}\text{-algebra isomorphism}
\end{aligned}$$

Quotient Modules and Module Hom

Def: Let R be ring. Let ${}_R M, {}_R N$ be modules

(1) $\phi: {}_R M \rightarrow {}_R N$ is called R -module hom if:

- (i) $\phi(x+y) = \phi(x) + \phi(y) \quad \forall x, y \in M$
- (ii) $\phi(rx) = r\phi(x) \quad \forall x \in M, r \in R$

$$\begin{aligned}
(2) \quad \ker \phi &= \{m \in {}_R M \mid \phi(m) = 0\} \\
\phi(M) &= \{n \in {}_R N \mid n = \phi(m) \text{ for some } m \in {}_R M\}
\end{aligned}$$

(3) $\text{Hom}_R(M, N)$ set of all module hom.

claim: $\ker \phi$ is an R -submodule of ${}_R M$.

Proof: $0 \in \ker \phi$

$\forall x, y \in \ker \phi, r \in R$

$$\phi(x+y) = \phi(x) + \phi(y) = 0$$

claim: $\phi(M)$ is an R -submodule of ${}_R N$

Def: $\phi(0) = 0 \in \phi(M)$

$\forall x \in \phi(M), r \in R$

$$\exists a, b \in M \quad x = \phi(a), y = \phi(b)$$

$$x + ry = \phi(a + rb) \in \phi(M)$$

Examples:

(1) ${}_R R$ R a ring, R -modules hom need not be ring hom
ring hom need not be R -module hom.

$$\begin{aligned}
\text{consider } \phi: {}_Z Z \rightarrow {}_Z Z \quad \begin{cases} \phi(n_1 + mn_2) = 2n_1 + 2mn_2 = \phi(n_1) + m\phi(n_2) \\ x \mapsto 2x \end{cases} \\
\Rightarrow \text{module hom.}
\end{aligned}$$

However, $x \mapsto 2x$ is not a ring hom $1 \mapsto 2$

consider $\phi: {}_F F[x] \rightarrow {}_F F[x]$

$$\phi(x) \mapsto \phi(x^2)$$

$$\begin{aligned}
\left[\begin{aligned} \phi(\phi(x) + \phi(x)) &= \phi(x^2) + \phi(x^2) = \phi(x^2) + \phi(x^2) \\ \phi(\phi(x)\phi(x)) &= \phi(x^2)\phi(x^2) = \phi(x^2)\phi(x^2) \end{aligned} \right] \\
\Rightarrow \text{ring hom.}
\end{aligned}$$

However, $x^2 = \phi(x) = x \cdot \phi(1) = x \cdot 1 = x$ but $\phi(x)$ is not a module hom.

(2) Let R be a ring, $n \in \mathbb{Z}^+, n \neq 1, M = R^n$

$$\pi_i: R^n \rightarrow R$$

$$(x_1, \dots, x_n) \mapsto x_i$$

$$\pi_i((x_1, \dots, x_n) + (y_1, \dots, y_n)) = \pi_i(x_1 + y_1, \dots, x_n + y_n) = x_i + y_i = \pi_i(x_1, \dots, x_n) + \pi_i(y_1, \dots, y_n)$$

surjective clearly.

$$\ker \pi_i = \{(x) \in R^n \mid x_i \in R, x_j = 0 \quad \forall j \neq i\}$$

(3) If R is a field, then R -modules are vector spaces $\Rightarrow$ module homs are linear transformations

(4) $R = \mathbb{Z}$ action on any $\mathbb{Z}$ -module is just addition and subtraction in abelian group structure.

$$\phi(2x) = \phi(x) + \phi(x) = 2\phi(x) \quad (\text{implied by (i)})$$

$\Rightarrow \mathbb{Z}$ -module homomorphisms are the same as abelian group homomorphisms.

(5) Let R be a ring, let I be a 2-sided ideal of R . Let ${}_R M, {}_R N$ be R -modules. Let $\text{ann}_R({}_R M) = \text{ann}_R({}_R N) = I$.

Then any $\phi: {}_R M \rightarrow {}_R N$ is an R/I module homomorphism.

If A is an additive group (abelian) let for some prime $p, px = 0 \quad \forall x \in A$.

Then any group homomorphism ψ is a $\mathbb{Z}/p\mathbb{Z}$ -module homomorphism (aka linear transformation) to itself.

$$(*) \text{Aut}(A) = \text{GL}(A)$$

Prop 2: Let ${}_R M, {}_R N, {}_R L$ be R -modules

(1) $\phi: {}_R M \rightarrow {}_R N$ is an R -module hom $\Leftrightarrow \phi(rx+y) = r\phi(x) + \phi(y) \quad \forall r \in R, \forall x, y \in M$

$$\text{Proof: (1)} \quad \phi(x+y) = \phi(r) = \phi(r+y) = \phi(rx) + \phi(y) \quad (\text{associative})$$

$$\Leftrightarrow \forall r=0, \phi(rx) = r\phi(x)$$

$$r=1, \phi(x+y) = \phi(x) + \phi(y)$$

(2) $\phi, \psi \in \text{Hom}_R(M, N)$

$$\text{Def: } (\phi + \psi)(m) = \phi(m) + \psi(m) \quad \forall m \in M$$

Then $\text{Hom}_R(M, N)$ is an abelian group

$$\begin{aligned}
\text{Def: } (i) \quad ((\phi + \psi) + \phi)(m) &= (\phi + \psi)(m) + \phi(m) \\
&= (\phi + (\psi + \phi))(m) \quad (N \text{ abelian}) \\
&= (\phi + \psi)(m) + \phi(m)
\end{aligned}$$

$$(ii) \quad 0 \in \text{Hom}_R(M, N)$$

$$(iii) \quad (\phi - \psi)(m) = \phi(m) - \psi(m) = 0 \quad \forall m \in M \quad (N \text{ abelian group})$$

$$(iv) \quad (\phi + \psi)(m) = \phi(m) + \psi(m) = \psi(m) + \phi(m) = (\psi + \phi)(m) \quad (N \text{ is abelian})$$

If R is commutative, then $\forall r \in R$

$$\text{Def: } (r\phi)(m) = r(\phi(m)) \quad \forall m \in M \Rightarrow r\phi \in \text{Hom}_R(M, N)$$

(i) $\text{Hom}_R(M, N)$ is additive abelian

(ii) Let $r, s \in R, \phi, \psi \in \text{Hom}_R(M, N)$

$$* \quad ((r+s)\phi)(m) = (r+s)(\phi(m)) = r(\phi(m)) + s(\phi(m)) = (r\phi)(m) + (s\phi)(m)$$

$$* \quad (rs)\phi(m) = (rs)(\phi(m))$$

$$= (sr)\phi(m)$$

$$= s(r\phi(m))$$

$$= s(\phi(rm))$$

If a has 1 , then

(3) If $\varphi \in \text{Hom}_R(L, M)$, $\psi \in \text{Hom}_R(M, N)$ then $\psi \circ \varphi \in \text{Hom}_R(L, N)$

Bsp: $\omega, l_1, l_2 \in \mathcal{L}$ $\omega, r \in R$

(4) $\text{Hom}_R(M, M)$ is a ring with $1 = \text{id}$ under $+$, $\circ$

$$\bullet ((\varphi_1 + \varphi_2) \circ \varphi_3)(m) = ((\varphi_1 + \varphi_2)(\varphi_3(m))) = \varphi_1 \circ \varphi_3(m) + \varphi_2 \circ \varphi_3(m) \quad \forall m \in \mathbb{N}.$$

$$\varphi \circ \text{id} = \text{id} \circ \varphi = \varphi$$

Def : $\text{Hom}_R(M, M) = \text{End}_R(M) = \text{End}(M)$.

If R has 1 , φ is a unital ring hom, and $\text{End}(M)$ is an $\mathcal{A}$ -algebra.
 $\varphi \mapsto \text{void}$ may not be injective as there may exist $r \in R$ s.t. $rm = 0 \forall m \in M$.

when R is a field, ϕ is injective, $R \cong \phi(R) \subseteq C(\text{End}(M))$ is called the copy of scalar transformations.

$$R \subseteq M/N \quad r(x+N) = (rx) + N \quad \forall r \in R, x+N \in M/N$$

Suppose $x + N = y + N \Rightarrow x - y \in N$
 $\Rightarrow r(x - y) \in N \quad \forall r \in R$
 $\Rightarrow rx - ry \in N$
 $\Rightarrow rx + N = ry + N$

$\Rightarrow$ axioms are satisfied (trivial)

$$\begin{aligned} (r+s)(x+N) &= (r+s)x + N = rx + sx + N = rx + N + sx + N = r(x+N) + s(x+N) \\ (rs)(x+N) &= (rs)x + N = r(sx) + N = r(s(x+N)) = r(s(x+N)) \\ r(x+N) + s(x+N) &= r(x+y+N) = rx + ry + N = r(x+N) + r(y+N) \end{aligned}$$

$\Rightarrow M/N$ is an $\mathbb{Q}$ -module

$\pi(x) = x + N$ is well defined. (abelian group here)

$$\pi(x+ry) = x+ry+N = x+N+r(y+N) = \pi(x) + r\pi(y) \Rightarrow \mathbb{R}\text{-module hom.}$$

Suppose $x \in \text{ker}(\pi)$. $\pi(x) = 0 + N = x + N \Rightarrow x \in N$

$$x \in N \quad \pi(x) = x + N = N = 0 + N$$

$$0 \in A + B$$

$$x, y \in A + B \quad r \in \mathbb{R} \Rightarrow x = a_1 + b_1, \quad y = a_2 + b_2 \quad a_1, a_2 \in A \quad b_1, b_2 \in B$$

$$x + ry = a_1 + b_1 + r(a_2 + b_2) = (a_1 + ra_2) + (b_1 + rb_2) \quad \text{ra}_2, +b_2 \in A, B \text{ respectively since } r \text{ is a scalar multiple.}$$

as $A + B$ is a submodule.

Thm 4 (1) Let $\phi \in \text{Hom}_{\mathbb{Z}}(M, N)$

$$N/\ker \varphi \cong \varphi(N) \text{ as additive abelian groups.}$$

- $\ker \varphi \subset_R M$, $\varphi(N) \subset_R N$ as k -submodules.

$$\phi: M/\ker \phi \longrightarrow \phi(M)$$

$$\phi(x + km\varphi) = \varphi(x)$$

$$\Rightarrow \phi(x+y+ze\lambda\varphi) = \phi(x+y) = \phi(x) + \phi(y) = \phi(x+ze\lambda\varphi) + \phi(y+ze\lambda\varphi)$$

$\Rightarrow \phi$ is an R -module isomorphism.

(2) $\varphi: A \longrightarrow A + B/R$ is a surjective abelian group hom.

$$a \mapsto a + \beta$$

$$q(a+rb) = a+rb + \mathfrak{B}$$

$$= a + b + r + s$$

$$+ \Delta + \delta + \tau(\delta + \delta))$$

$$= \varphi(a) + r\varphi(b) \Rightarrow \text{surjective } \mathfrak{A}\text{-module hom.}$$

$\ker \varphi = A \cap B \Rightarrow \text{by (1)} \quad A/A \cap B \cong A+B/B, \text{ as } I\text{-modules.}$

(3) $A \subset B$ submodules of M .

claim: $\mathbb{Z}/A \subset M/A$ as submodule.

Ex: $E/A \subset N/A$ as abelian subgroup.

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isNotEmpty()
```

Let $b_1 + A, b_2 + A \in \mathbb{R}/A$ with $r \in \mathbb{R}, b_1, b_2 \in B$

$$b_1 + A + r b_2 + A = (b_1 + r b_2) + A \in B/A$$

$$\text{Ex } \varphi: M/A \longrightarrow M/B$$

$$x+A \mapsto x+B \quad \text{is an surjective abelian group hom.}$$

$$Q(x + A + r(y + A)) = Q(x + ry + A) = x + ry + B = x + B + ry + B$$

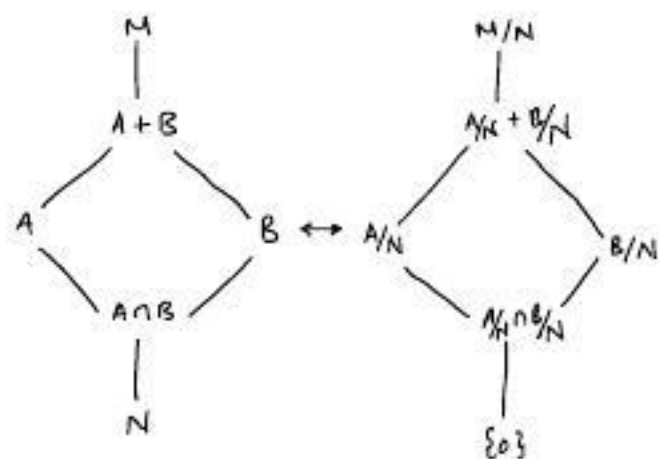
$$= \varphi(x+A) + \gamma \varphi(y+A)$$

$\Rightarrow$ A -module, n.g.m.

$$R[A] \cong B/A \Rightarrow (M/A)/(N/A) \cong M/N \text{ as } R \text{ modules}$$

(4) $W = NC = N$

$\{A\} \hookrightarrow \{A/N\} \quad \forall A \in \mathcal{N}$ as R -submodules



Exercises

R be a ring with 1. M is a left R -module.

2. $\sim$ isomorphic to on set of R -modules $\mathcal{M}$

with $A, B, C \in \mathcal{M}$

(i) $A \cong A$ as R -module by $\text{id} \in \text{Hom}_R(A, A)$

(ii) $A \cong B \Rightarrow \exists \varphi: B \rightarrow A \cong B \cong A$

(iii) $A \cong B, B \cong C, \varphi: A \rightarrow B, \psi: B \rightarrow C$

$\Rightarrow \psi \circ \varphi: A \rightarrow C$ is an isomorphism (3)

$\Rightarrow A \cong C$

$\Rightarrow \cong$ is an equivalence relation.

(3) $R = \mathbb{F}[X], M = \mathbb{F}[X], R^M$

$\varphi: f(x) \mapsto f(x^2)$ is a ring homomorphism ^{addition} thus a group automorphism. However, it is not an R -module homomorphism. $[\varphi(x) = x \cdot \varphi(1) = x]$

(4) Let A be any $\mathbb{Z}$ -module, $a \in A, n \in \mathbb{Z}^+$

$$\varphi_a: \mathbb{Z}/n\mathbb{Z} \rightarrow A$$

$$\bar{x} \mapsto xa$$

(i) Suppose φ_a is a well defined $\mathbb{Z}$ -module hom.

$$\Rightarrow 0 = \varphi_a(\bar{0}) = \varphi_a(n \cdot \bar{1}) = n \cdot \varphi_a(\bar{1}) = na$$

(ii) Suppose $na = 0$

$$\bar{x} = \bar{y} \text{ in } \mathbb{Z}/n\mathbb{Z}$$

$$\bar{x} - \bar{y} = \bar{0} \text{ in } \mathbb{Z}/n\mathbb{Z} \Rightarrow x - y = mn, m \in \mathbb{Z}$$

$$0 = mn a = \varphi_a(\bar{x} - \bar{y}) = (x - y)a = xa - ya = \varphi_a(\bar{x}) - \varphi_a(\bar{y}) \Rightarrow \varphi_a(\bar{x}) = \varphi_a(\bar{y}) \quad [A \text{ is a } \mathbb{Z}\text{-module}]$$

$\Rightarrow$ well defined.

Suppose $\bar{x}, \bar{y} \in \mathbb{Z}/n\mathbb{Z}$, $m \in \mathbb{Z}$

$$\varphi_a(\bar{x} + m\bar{y}) = (x + my)a = xa + mya$$

$$= \varphi_a(\bar{x}) + m\varphi_a(\bar{y}) \Rightarrow \mathbb{Z}\text{-module homomorphism.}$$

$$\phi: A_n \rightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$$

$$a \mapsto \varphi_a$$

$$\text{if } a = b \in A_n$$

$$\phi(a) = \varphi_a(\bar{x}) = xa = kb = \varphi_b(\bar{x}) = \phi(b) \quad \forall \bar{x} \in \mathbb{Z}/n\mathbb{Z} \quad [\varphi_a, \varphi_b \text{ are } \mathbb{Z}\text{-module hom. since } a, b \in A_n]$$

$\Rightarrow$ well defined.

Suppose $a, b \in A_n, m \in \mathbb{Z}, \bar{x} \in \mathbb{Z}/n\mathbb{Z}$

$$\phi(a + mb)(\bar{x}) = \varphi_{a+mb}(\bar{x}) = (a + mb)x = ax + mbx = \varphi_a(\bar{x}) + m\varphi_b(\bar{x}) = \phi(a)(\bar{x}) + m\phi(b)(\bar{x})$$

$$\forall \bar{x} \in \mathbb{Z}/n\mathbb{Z}$$

$\Rightarrow \phi$ is a $\mathbb{Z}$ -module-hom ($\mathbb{Z}$ -commutative)

Suppose $x \in \ker \phi$

$$\phi(x)(\bar{x}) = \varphi_x(\bar{x}) = x\bar{x} = 0 \quad \forall \bar{x} \in \mathbb{Z}/n\mathbb{Z} \Rightarrow x = 0$$

Suppose $\tilde{\phi} \in \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A_n)$

Then $\forall \bar{x} \in \mathbb{Z}/n\mathbb{Z}$

$$0 = \tilde{\phi}(n \cdot \bar{x}) = n \cdot \tilde{\phi}(\bar{x}) \Rightarrow \tilde{\phi}(\bar{x}) \in A_n$$

$$\Rightarrow \tilde{\phi}(\bar{1}) = b \in A_n$$

Since ϕ is well-defined, $\phi(b) = \tilde{\phi} = \varphi_b \Rightarrow$ Thus, ϕ is surjective

and $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong A_n = \text{Ann}_A((n))$

$$(5) \text{ Hom}_{\mathbb{Z}}(\mathbb{Z}/30\mathbb{Z}, \mathbb{Z}/21\mathbb{Z}) \cong \text{Ann}_{\mathbb{Z}/21\mathbb{Z}}((30)) = \{\bar{a} \in \mathbb{Z}/21\mathbb{Z} \mid 30 \cdot \bar{a} = \bar{0} \text{ in } \mathbb{Z}/21\mathbb{Z}\} = \langle \bar{7} \rangle = \{0, 7, 14\} \cong \mathbb{Z}/3\mathbb{Z}$$

$$(6) \text{ Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \text{Ann}_{\mathbb{Z}/n\mathbb{Z}}((m))$$

$$\Rightarrow m \cdot \bar{a} = \bar{0} \text{ in } \mathbb{Z}/n\mathbb{Z}$$

$$\Rightarrow \bar{a} \text{ has order } \gcd(m, n) \text{ in } \mathbb{Z}/n\mathbb{Z}$$

$$\Rightarrow \text{Ann}_{\mathbb{Z}/n\mathbb{Z}}((m)) \text{ is a subgroup of } \mathbb{Z}/n\mathbb{Z} \text{ of order } \gcd(m, n)$$

$$\Rightarrow \text{Ann}_{\mathbb{Z}/n\mathbb{Z}}((m)) \cong \mathbb{Z}/\gcd(m, n)\mathbb{Z}$$

(7) $z \in C(R)$ is fixed.

$$m \mapsto zm \in \text{Hom}_R(M, M)$$

$$m + rn \mapsto zm + zr n = zm + rzn \quad \forall m, n \in M, r \in R$$

Suppose R commutative. $R \rightarrow \text{End}_R(M) \quad r \mapsto rI$

$$r_1 + r_2 \mapsto (r_1 + r_2)I = r_1I + r_2I$$

$$r_1 r_2 \mapsto (r_1 r_2)I = r_1(r_2I)$$

$$m \in M \quad (r_1(r_2I))(m)$$

$$\begin{aligned}
 &= r_1(r_2 I(m)) \\
 &= r_1(I(r_2 m)) \\
 &= r_1 I(I(r_2 m)) \\
 &= r_1 I(r_2 I(m)) = r_1 I \circ r_2 I.
 \end{aligned}$$

$\Rightarrow r \mapsto rI$ is a ring hom.

8) Let $\varphi \in \text{Hom}_R(N, N)$

Suppose $n \in \varphi(\text{Tor}(M))$

$$\exists a \in \text{Tor}(M) \text{ s.t. } n = \varphi(a)$$

$$\exists r \in R \text{ s.t. } ra = 0$$

$$\Rightarrow rn = r\varphi(a) = \varphi(ra) = 0 \Rightarrow n \in \text{Tor}(N) \Rightarrow \varphi(\text{Tor}(M)) \subseteq \text{Tor}(N)$$

9) Let R be commutative.

$\text{Hom}_R(R, M) \cong M$ as left R -modules.

$$\varphi \in \text{Hom}_R(R, M)$$

$$\exists \varphi_1, \varphi_2 \in \text{Hom}_R(R, M)$$

$$\exists \varphi_1(z) = \varphi_2(z)$$

$$\varphi_1(r) = r \cdot \varphi_1(1) = r \cdot \varphi_2(1) = \varphi_2(r) \quad \forall r \in R.$$

$$\Rightarrow \varphi_1 = \varphi_2 \text{ in } \text{Hom}_R(R, M)$$

$$\text{Def } \phi: \text{Hom}_R(R, M) \rightarrow M$$

$$\varphi \mapsto \varphi(1)$$

$$\varphi = \psi \text{ in } \text{Hom}_R(R, M) \text{ then clearly } \varphi(1) = \psi(1) \quad \forall \varphi \in R$$

since, $\phi(1) = \psi(1) \Rightarrow$ well defined.

$$\phi(\varphi + \tau\psi) = (\varphi + \tau\psi)(1)$$

$$= \varphi(1) + (\tau\psi)(1)$$

$$= \varphi(1) + \tau(\psi(1))$$

$$= \phi(\varphi) + \tau\phi(\psi) \Rightarrow \phi \text{ is an } R\text{-module hom.}$$

Suppose $\varphi \in \ker \phi$

$$0 = \phi(\varphi) = \varphi(1)$$

$$\forall r \in R \quad \varphi(r) = r\varphi(1) = 0 \Rightarrow \varphi = 0.$$

Suppose $m \in M$.

$$\text{Let } \varphi: R \rightarrow M$$

$$\text{s.t. } \varphi(1) = m$$

$$r_1 = r_2 \quad \varphi(r_1) = r_1 m = r_2 m = \varphi(r_2) \text{ [well def.]}$$

$$\varphi(r_1 + r_2) = (r_1 + r_2)m = r_1 m + r_2 m = \varphi(r_1) + \varphi(r_2) \Rightarrow R\text{-module hom.}$$

$$\Rightarrow \varphi \in \text{Hom}_R(R, M)$$

$$\text{In particular, } \varphi(1) = m \Rightarrow \phi(\varphi) = m.$$

$$\Rightarrow \phi \text{ is surjective.}$$

$$\Rightarrow \phi \text{ is an isomorphism. } \Rightarrow \text{Hom}_R(R, M) \cong M \text{ as left } R\text{-modules.}$$

10) $\text{Hom}_R(R, R) \cong R$ as left R -modules. $\text{Hom}_R(R, R)$ is a ring with 1.

Consider

$$\phi: \varphi \mapsto \varphi(1)$$

$$\phi(\varphi \circ \psi) = \varphi \circ \psi(1) = \varphi(r) = \varphi(r \cdot 1) = r \cdot \varphi(1) = \psi(1) \cdot \varphi(1) \text{ [R commutative]}$$

$$\Rightarrow \text{ring isomorphism.}$$

$$\Rightarrow \text{Hom}_R(R, R) \cong R \text{ as rings.}$$

11) $A_1, \dots, A_n$ are R -modules. $B_i \subseteq A_i$ submodules $\forall i \in \{1, 2, \dots, n\}$.

$$\pi: A_1 \times \dots \times A_n \rightarrow A_1/B_1 \times \dots \times A_n/B_n$$

$$(a_1, \dots, a_n) \mapsto (\pi_1(a_1), \dots, \pi_n(a_n))$$

$$\pi_i: A_i \rightarrow A_i/B_i \text{ quotient } R\text{-module hom.}$$

π is an R -module hom.

$$\ker(\pi) = B_1 \times \dots \times B_n \text{ submodule.}$$

$$\Rightarrow (A_1 \times \dots \times A_n) / (B_1 \times \dots \times B_n) \cong (A_1/B_1) \times \dots \times (A_n/B_n)$$

12) Suppose I left ideal of R

IR is a submodule of R as R -module.

$$(R \times \dots \times R) / (IR \times IR \times IR \dots) \cong R/IR \times \dots \times R/IR \text{ } n \text{ times.}$$

$$[IR^n = IR \times IR \times IR \dots \times IR \text{ } n \text{ times}]$$

13) I is a nilpotent ideal $[\exists n \in \mathbb{N} \text{ s.t. } I^n = 0]$

M, N are R -modules.

Let $\varphi \in \text{Hom}_R(M, N)$

Suppose $\bar{\varphi}: M/IM \rightarrow N/IN$ is surjective.

$$\varphi|_{I^{n-1}M}: I^{n-1}M \rightarrow I^{n-1}N$$

$$y \in I^{n-1}N \quad y = \sum a_j n_j \quad a_j \in I^{n-1} \quad n_j \in N$$

$$n_j \equiv \varphi(m_j) \pmod{IN} = \varphi(m_j) + z_j \quad z_j \in IN$$

$$y = \varphi(\sum a_j m_j) + \sum a_j z_j = \varphi(\sum a_j m_j) \quad a_j z_j \in I^n N = \{0\}$$

$\Rightarrow$ it is surjective.

Suppose $\varphi|_{I^{n-k}M}: I^{n-k}M \rightarrow I^{n-k}N$ is surjective for some $n-1 > k \geq 1$.

$$\text{Then consider } \varphi|_{I^{n-(k+1)}M}: I^{n-(k+1)}M \rightarrow I^{n-(k+1)}N$$

$$\text{Let } y \in I^{n-(k+1)}N \quad y = \sum a_j n_j \quad a_j \in I^{n-(k+1)} \quad n_j \in N$$

$$n_j \equiv \varphi(m_j) \pmod{\langle \mathbb{N} \rangle} = \varphi(m'_j) + z_j \quad z_j \in \mathbb{N}$$

$$\text{Then, } y = \varphi(E a_j m_j) + a_j z_j$$

$$a_j z_j \in I^{n-k} N \Rightarrow \exists m' \in I^{n-k} N \text{ s.t. } \varphi(m') = a_j z_j$$

$$\Rightarrow y = \varphi(E a_j m_j + m')$$

$\Rightarrow$ it is surjective.

$\Rightarrow \varphi|_{IN}$ is surjective

$$\text{Suppose } n \in \mathbb{N} \quad n = \varphi(m) + z_j \quad z_j \in \mathbb{N}$$

$$\exists m' \in IN \text{ s.t. } \varphi(m') = z_j$$

$$\Rightarrow n = \varphi(m + m') \Rightarrow \varphi \text{ is surjective.}$$

$$(14). R = \mathbb{Z}[x] \quad \text{let } A = \mathbb{Z}[t_1, t_2, \dots]$$

$$\text{Def } R \supset A$$

$$1) 1 \cdot p = p \quad \forall p \in A$$

$$2) n \geq 1 \quad x^n \cdot 1 = t_n$$

$$x^n \cdot t_i = t_{n+i} \quad \text{for } i=1, 2, 3, \dots$$

$$x^0 \cdot t_i = 0 \quad \text{if } i \geq 2 \quad t_i = t_1^i, t_2^i, \dots$$

$$3) \forall r(x), s(x) \in R \quad \forall p, q \in A$$

$$(r(x) + s(x)) \cdot p = r(x) \cdot p + s(x) \cdot p$$

$$r(x) \cdot (p + q) = r(x) \cdot p + r(x) \cdot q$$

$$(14) \quad x^{r+s} \cdot t_i = t_{r+s+i} = t_{r+(s+i)} = x^r \cdot t_{s+i} = x^r \cdot (x^s \cdot t_i)$$

$$\text{let } r, s \in \mathbb{Z}[x], p \in A$$

$$(rs) \cdot p = \sum_{i,j} a_i b_j (x^{i+j} \cdot p)$$

$$= \sum_{i,j} a_i b_j (x^i \cdot (x^j \cdot p))$$

$$= \sum_i a_i x^i \cdot \left(\sum_j b_j x^j \cdot p \right)$$

consider $M \in A$ monomial of $\deg \geq 2$

then $x^{i+j} \cdot M = x^i \cdot (x^j \cdot M) = 0$, we already prove for degree 1 monomials and it is clearly true for integers.

$\Rightarrow A$ is a $\mathbb{Z}[x]$ module. (unital)

$$(b) \quad \varphi: R \rightarrow A \quad \varphi(r) = r \circ 1_A$$

$$\text{let } r_1(x), r_2(x), r'(x) \in R$$

$$\begin{aligned} \varphi(r_1(x) + r'(x)r_2(x)) &= (r_1(x) \circ 1_A) + r'(x) \cdot (r_2(x) \circ 1_A) \\ &= \varphi(r_1(x)) + r'(x) \cdot \varphi(r_2(x) \circ 1_A) \end{aligned}$$

$$\varphi(1_R) = 1_R \circ 1_A = 1_A \Rightarrow \text{is an } R\text{-module hom.}$$

$$\begin{aligned} \varphi(x^n) \cdot \varphi(x^m) &= (x^n \circ 1_A) \cdot (x^m \circ 1_A) \\ &= t_n \cdot t_m \end{aligned}$$

$$\text{However, } \varphi(x^{n+m}) = x^{n+m} \circ 1_A = t_{n+m} \Rightarrow \text{thus it is not a ring hom.}$$

10.3 Generation of modules, direct sums and free modules

let R be a ring with 1.

Def let M be an R -module. let $N_1, \dots, N_n$ be submodules.

$$(1) \quad N_1 + \dots + N_n := \{a_1 + \dots + a_n \mid a_i \in N_i \forall i\}$$

$$(2) \quad \forall A \subset M \text{ subset}$$

$$RA := \{r_1 a_1 + r_2 a_2 + \dots + r_m a_m \mid r_1, \dots, r_m \in R, a_1, \dots, a_m \in A, m \in \mathbb{Z}^+\} \text{ (submodule generated by } A \text{)}$$

$$[RA = \{0\} \text{ if } A = \emptyset]$$

If $N \subseteq M$ is a submodule and $N = RA$, for some subset A of M , we say N is generated by A .

(3) submodule $N \subseteq M$ is finitely generated if $\exists$ some finite subset A of M s.t. $N = RA$.

(4) submodule $N \subseteq M$ is cyclic if $\exists a \in M$ s.t. $N = Ra$

• RA is a submodule of M . $0 \in RA$

$$\sum r_i a_i, \sum s_i b_i \in RA \quad r_i \in R \text{ (finite sums)}$$

$$\sum r_i a_i + r \sum s_i b_i = \sum r_i a_i + \sum r s_i b_i \in RA$$

suppose $A \subset N$ a subset of M .

$$\text{let } \sum r_i a_i \in RA$$

$$\forall i \quad r_i a_i \in N \Rightarrow \sum r_i a_i \in N \Rightarrow RA \subseteq N$$

In particular, $N_1, \dots, N_n$ submodules of M .

$N_1 + N_2 + \dots + N_n$ is generated by $N_1 \cup N_2 \cup \dots \cup N_n$

and it is the smallest submodule containing $N_i \forall i$.

If N_i are generated by sets $A_i \subset M$.

Then, $N_1 + N_2 + \dots + N_n = R(A_1 \cup A_2 \cup \dots \cup A_n)$

A submodule N of M may have different generating sets.

for example, $N = RN$

Also, $N = RA$

If N is finitely generated, $\exists d \in \mathbb{Z}^+$ s.t. $N = RA$ $|A| = d$
 $\forall B \subset M$ s.t. $|B| < d$ $N \neq RB$.

subset A is called a minimal set of generators. (not unique in general)

If N is not a finitely generated submodule, it does not have a minimal generating set.

Examples

(1) $R = \mathbb{Z}$ M any R -module.

If $a \in M$ $\mathbb{Z}a = \langle a \rangle$

M is generated as a $\mathbb{Z}$ -module by set $A \Leftrightarrow M$ is generated by A as abelian group.

(2) let R be a ring with 1. let $M = R^R$

$R^R = R\{1\}$ finitely generated.

if $I = Ra \Leftrightarrow I$ is a principal ideal of R .

if I is finitely generated R -module of $R = I$ is a finitely generated ideal.

When R is commutative, we write aR or RA for submodule(ideal) generated by a or A resp.

$aR = Ra$ and $aR = Ra$. Thus, P.I.D is an integral domain R with 1 in which every R -submodule of R is cyclic.

* submodules of a finitely generated module need not be finitely generated.

eg. $M = R^R$ $R = F[t_1, t_2, t_3, \dots]$

$R\{t_1, t_2, \dots\}$ cannot be generated by a finite set.

suppose $I \subset R\{t_1, t_2, \dots\}$ is a finite set. that generates $R\{t_1, t_2, \dots\}$.

$R\{t_1, t_2, \dots\} = RI$ $\{t_1, \dots\} \subset I \nsubseteq$

(3) let R be a ring with 1.

let $M = R^n$

$\forall i \in \{1, 2, \dots, n\}$ let $e_i = (0, \dots, 0, \underset{i}{1}, 0, \dots, 0)$

$(s_1, s_2, \dots, s_n) = \sum_{i=1}^n s_i e_i \Rightarrow M = R\{e_1, \dots, e_n\}$

(4) consider $F[x]$.

let V be a vector space over F

let $T \in \text{End}(V)$

* V is a cyclic $F[x]$ -module (with generator v) $\Leftrightarrow V = \{p(x)v \mid p(x) \in F[x]\}$
 $\Leftrightarrow V = \text{span}\{T^n(v) \mid n \geq 0\}$

$\Rightarrow V$ is a cyclic $F[x]$ -module $V = F[x][v]$.

$u \in V$ $u = p(x)v$ for some $p(x) \in F[x]$.

$u = a_n T^n v + \dots + a_0 v \Rightarrow u \in \text{span}\{T^n(v) \mid n \geq 0\}$

$\Leftrightarrow V = \text{span}\{T^n(v) \mid n \geq 0\} \Rightarrow V = \text{span}\{T^n(v) \mid n \geq 0\}$

$\forall v' \in V$ $v' = p(T)v = p(x) \cdot v$

$V = F[x][v]$.

If $T = I$ or $T = 0$

$\forall v \in V$ $\forall p \in F[x]$ $p(x)v = av$ for some $a \in F$

if $\dim V \geq 1$, V cannot be a cyclic $F[x]$ -module.

suppose $V = F^n$ T shift operator.

$T(e_n) = e_{n-1}$ for $1 \leq n \leq n$

$V = \text{span}(e_n, T(e_n), \dots, T^{n-1}(e_n)) \Rightarrow V$ is a cyclic $F[x]$ -module with generator e_n .

For, $n \geq 1$, V is not a cyclic F -module.

Def let $\{M_1, \dots, M_k\}$ be a collection of R -modules

The collection $(m_1, m_2, \dots, m_k)$ $m_i \in M_i$ with addition and action of R defined componentwise, is called direct product of $M_1, \dots, M_k = M_1 \times \dots \times M_k$

$M_1 \times \dots \times M_k$ is clearly an R -module. $= M_1 \oplus M_2 \oplus \dots \oplus M_k$

Prop 3 let $N_1, \dots, N_k$ be submodules of R -module M . Then following are equivalent:

(1) $\pi: N_1 \times \dots \times N_k \rightarrow N_1 + N_2 + \dots + N_k$

$\pi(a_1, \dots, a_k) = a_1 + \dots + a_k$

is an isomorphism: $N_1 + N_2 + \dots + N_k \cong N_1 \times \dots \times N_k$

(2)

(2) $N_j \cap (N_1 + N_2 + \dots + N_{j-1} + N_{j+1} + \dots + N_k) = 0 \quad \forall j \in \{1, 2, \dots, k\}$
 (3) $\forall x \in N_1 + \dots + N_k$ can be uniquely written in the form $a_1 + a_2 + \dots + a_k$ with $a_i \in N_i$

Proof (1) $\Rightarrow$ (2)

Suppose $a \in N_j \cap (N_1 + N_2 + \dots + \hat{N}_j + \dots + N_k)$, $a \neq 0$
 $a = n_j = n_1 + n_2 + \dots + \hat{n}_j + \dots + n_k$
 $n_j = n_1 + n_2 + \dots + \hat{n}_j + \dots + n_k \Rightarrow \pi(n_1, \dots, -a, \dots, -n_k) = 0 \quad \nexists$

(2) $\Rightarrow$ (3)

Suppose $x = a_1 + \dots + a_k = b_1 + \dots + b_k \quad a_i, b_i \in N_i$

$\forall i \in \{1, 2, \dots, k\}$

$a_i - b_i = (b_1 - a_1) + \dots + (b_k - a_k) \in N_i \cap (N_1 + \dots + \hat{N}_i + \dots + N_k) = 0$

$\Rightarrow a_i - b_i = 0 \Rightarrow a_i = b_i$

(3) $\Rightarrow$ (1)

π is clearly well defined.

π is clearly surjective.

π is an R -module homomorphism

by (2) it is injective.

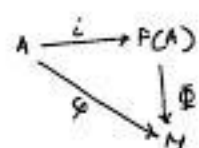
If an R -module $M = N_1 + \dots + N_k$ $N_1, \dots, N_k \subset M$ submodules

satisfying prop 6 then $M = N_1 \oplus \dots \oplus N_k$ (internal direct sum)

$\forall m \in M \quad m = n_1 + \dots + n_k$ uniquely $n_i \in N_i$

Def R -module F is said to be free on the subset A of F if $\forall x \neq 0 \quad x \in F$, $\exists$ nonzero $r_1, \dots, r_n \in R$ and $a_1, \dots, a_n \in A$ s.t. $x = r_1 a_1 + \dots + r_n a_n$ for some $n \in \mathbb{Z}^+$
 In this case, A is a basis for F .

Theorem 6 For any set A there is a free R -module $F(A)$ on the set A and $F(A)$ satisfies following universal property: If M is any R -module and $\varphi: A \rightarrow M$ any map of sets, then $\exists$ a unique R -module hom $\Phi: F(A) \rightarrow M$ s.t. $\Phi(a) = \varphi(a) \quad \forall a \in A$.



When A is a finite set $\{a_1, \dots, a_n\}$, $F(A) = Ra_1 \oplus Ra_2 \oplus \dots \oplus Ra_n \cong R^n$

Def: Let $F(A) = 0$ if $A = \emptyset$. If $A \neq \emptyset$, w $F(A) = \{f: A \rightarrow R \mid f(a) = 0 \text{ for all but finitely many } a \in A\}$

Def: $(f+g)(a) = f(a) + g(a)$

$(rf)(a) = r(f(a)) \quad \forall a \in A, r \in R, f, g \in F(A)$

check:

$F(A)$ abelian group under addition

$\bullet ((f+g)+h)(a) = (f+g)(a) + h(a) = f(a) + g(a) + h(a) = (f+(g+h))(a)$

$\bullet (f+0)(a) = f(a) = (0+f)(a)$

$\bullet (f+(-1)f)(a) = f(a) + (-1)f(a) = f(a) - f(a) = 0$

$R \ni F(A)$:

w $r_1, r_2 \in R, f, g \in F(A) \quad a \in A$

$\bullet ((r_1+r_2)f)(a) = (r_1+r_2)(f(a)) = r_1 f(a) + r_2 f(a) = (r_1 f + r_2 f)(a)$

$\bullet ((r_1 r_2)f)(a) = r_1 r_2 (f(a)) = r_1 (r_2 f(a)) = (r_1 (r_2 f))(a)$

$\bullet (r_1(f+g))(a) = r_1((f+g)(a)) = r_1(f(a) + g(a)) = r_1 f(a) + r_1 g(a) = (r_1 f + r_1 g)(a)$

$\Rightarrow F(A)$ is an R -module

$i: A \rightarrow F(A)$

$a \mapsto f_a \quad f_a(x) = \begin{cases} 1 & x=a \\ 0 & x \neq a \end{cases}$

$\forall a=b \quad f_a(a) = f_a(b) = 1 = f_b(a) = f_b(b)$

zero everywhere else

$\Rightarrow f_a = f_b \Rightarrow$ well defined

Then every $f \in F(A) \quad f = r_1 f_{a_1} + \dots + r_n f_{a_n} = r_1 a_1 + \dots + r_n a_n$ uniquely.
 $f(a_i) = r_i f_{a_i}(a_i) = r_i$

Def $\Phi: F(A) \rightarrow M$

$r_1 a_1 + \dots + r_n a_n \mapsto r_1 \varphi(a_1) + \dots + r_n \varphi(a_n)$

$\sum_i r_i a_i = \sum_j s_j b_j$ in $F(A) \Rightarrow i=j$ and $r_i = s_i$ by uniqueness.
 $a_i = b_i$

Then $\Phi(\sum_i r_i a_i) = \Phi(\sum_i s_i b_i)$

Suppose $x, y \in F(A) \quad r \in R$

$\Phi(x+ry) = \Phi(\sum_i r_i a_i + r \sum_j s_j b_j) = \sum_i r_i \varphi(a_i) + \sum_j r s_j \varphi(b_j)$
 $= \Phi(x) + r \Phi(y)$

$\Rightarrow \Phi$ is an R -module homomorphism.

Since $F(A) = R\{A\} \Rightarrow$ we can determine uniquely the values of Φ on $F(A)$ by looking values it takes on $\{A\} \Rightarrow \Phi$ is unique extension of φ to all of $F(A)$.

If A is finite set $\{a_1, \dots, a_n\}$ then $F(A) = Ra_1 \oplus \dots \oplus Ra_n \cong Ra_1 \times Ra_2 \times \dots \times Ra_n$

$$ra_i \in R \quad \forall i \in \{1, 2, \dots, n\}$$

$$(ra_i \mapsto r) \Rightarrow F(A) \cong R^n$$

Corollary 7

(1) If F_1, F_2 are free modules on the same set A . $\exists$ unique isomorphism between F_1 and F_2 , which is the identity map on A .

Proof:

$$\begin{array}{ccc} A & \xrightarrow{i_1} & F_1(A) \\ & \searrow i_2 & \downarrow \Phi \\ & & F_2(A) \end{array}$$

$$\Phi(\sum r_i a_i) = \sum r_i i_2(a_i) \text{ clearly an isomorphism.}$$

$$\Phi(\sum r_i a_i) = 0 = \sum r_i i_2(a_i) = 0 \Rightarrow r_i = 0 \quad \forall i$$

$$\Rightarrow \sum r_i a_i = 0 \text{ in } F_1(A) \Rightarrow \ker \Phi = \{0\}.$$

$$\sum r_i i_2(a_i) = \sum r_i \Phi(a_i)$$

$$= \Phi(\sum r_i a_i) \Rightarrow \Phi \text{ is an isomorphism. } \Phi|_A : F_1(A) \rightarrow F_2(A) \quad \Phi(a_i) = i_2(a_i) \text{ thus, identity on } A.$$

(2) If $F = R\{A\}$, by (1) $F \cong F(A)$.

$$\begin{array}{ccc} A & \xrightarrow{i} & F \\ & \searrow i & \downarrow \Phi \\ & & F(A) \end{array}$$

If F is a free R -module with basis A , we can define R -modules home from F into other R -modules by simply specifying their values on the elements of A and then simply extend by linearity.

When $R = \mathbb{Z}$, free module on a set A is called the free abelian group on A .

If $|A| = n$ then $F(A) \cong \mathbb{Z} \oplus \dots \oplus \mathbb{Z}$ (n times) is called the free group of rank n .

Exercises R ring with 1, M is left R -module

(1) Suppose $|A| = |B| = n$ A, B two sets.

Let $\phi: A \rightarrow B$ is a bijection.

$$a_i \mapsto b_i$$

$$\begin{array}{ccc} A & \xrightarrow{i} & F(A) \\ \downarrow \phi & & \downarrow \Phi \\ B & \xrightarrow{i} & F(B) \end{array}$$

$\phi \circ \phi: A \rightarrow F(B)$ is a well defined map.

$\exists! \Phi$ R -module hom. $\in \text{Hom}_R(F(A), F(B))$ $\Phi(a_i) = \phi(a_i)$

Suppose $a \in \ker \Phi$ $\Phi(a) = \Phi(\sum r_i a_i) = \sum r_i \phi(a_i) = 0 \Rightarrow r_i = 0 \Rightarrow a = 0$.

Let $b \in F(B)$ $b = \sum r_i b_i$ since $\phi: A \rightarrow B$ bijective, $\Rightarrow \sum r_i b_i = \sum r_i \phi(a_i) = \Phi(\sum r_i a_i)$

(2) Suppose R commutative with 1. $R^n \cong R^m$

$\Rightarrow$ surjective and $F(A) \cong F(B)$ as modules.

Suppose I is maximal ideal of R (R/I field)

$$R^n / IR^n \cong R/I \times \dots \times R/I \text{ (n times)}$$

is R/I annihilated by I ?

$$r + IR \quad \text{let } r \in I$$

$$a(r + IR) = ar + IR = 0 + IR$$

$$\Rightarrow R/IR \text{ is an } R/I\text{-module. } \Rightarrow R/IR \text{ is a vector space over } R/I$$

$$\Rightarrow R^n / IR^n \cong (R/IR)^n \cong (R/IR)^m \text{ (as } m=n \text{ vector spaces).}$$

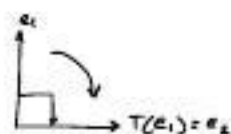
(3) $F = R$ $V = R^2$

$T: V \rightarrow V$ rotation $-\pi/2$ by origin.

$$\Rightarrow e_1 = (0, 1) \quad T(e_1) = (1, 0)$$

$$V = R^2 = \text{span}\{e_1, e_2\} \Rightarrow V = F[x]\{e_1\}$$

$\Rightarrow V$ is a cyclic module.



$$F = R$$

$$T: V \rightarrow V \quad (x, y) \mapsto (y, -x)$$

consider $v = (1, 1)$

$$T(v) = (1, -1)$$

$$V = R^2 = \text{span}\{(1, 1), (1, -1)\} \Rightarrow V = F[x]\{(1, 1)\} \Rightarrow V \text{ is a cyclic module.}$$

(4) An R -module M is called torsion module if $\forall m \in M \quad \exists r \neq 0 \in R$ s.t. $rm = 0$, where r may depend on the choice of m .

Suppose A is a finite abelian group $\Rightarrow A$ is a $\mathbb{Z}$ -module.

$$\forall a \in A \quad 1 \cdot a = 0 \quad 1 \mid n \Rightarrow \text{Tor}(A) = A.$$

$M = \mathbb{Z}/p\mathbb{Z}[x]$ as a $\mathbb{Z}$ -module. $\mathbb{Z}/p\mathbb{Z}$ is an infinite abelian group.

$$\forall f(x) \in \mathbb{Z}/p\mathbb{Z}[x] \quad p \cdot f(x) = 0 \Rightarrow \text{Tor}(\mathbb{Z}/p\mathbb{Z}[x]) = \mathbb{Z}/p\mathbb{Z}[x].$$

(5) Let R be an integral domain.

Suppose A is a finite set that generates M i.e. $M = R\{a_1, \dots, a_n\} = \text{Tor}(M)$.

$$\forall a_i \quad \exists r_i \text{ s.t. } r_i a_i = 0$$

Suppose $m \in M$ $m = s_1 a_1 + \dots + s_n a_n$

$$\text{Then } (r_1, \dots, r_n)m = (r_1, \dots, r_n)r_1a_1 + \dots + (r_1, \dots, r_n)s_na_n = 0$$

$$\Rightarrow \forall m \in M, (r_1, \dots, r_n)m = 0.$$

$$R \cong \mathbb{Z}, M = \mathbb{Z}$$

M/N is a $\mathbb{Z}$ -module.

$$\forall \frac{q}{b} + \mathbb{Z} \text{ in } M, \quad b \cdot \frac{q}{a} + \mathbb{Z} = 0 + \mathbb{Z} \Rightarrow \text{Tor}(M) = M.$$

$$\text{Suppose } \text{Ann}_{\mathbb{Z}}(M) \neq \{0\}. \quad \exists n \in \mathbb{Z}, n \neq 0, \text{ s.t. } n(q + \mathbb{Z}) = 0 \quad \forall q + \mathbb{Z} \in M$$

$$\text{However, } n\left(\frac{1}{n+1} + \mathbb{Z}\right) \neq 0 \text{ since } \frac{n}{n+1} \notin \mathbb{Z}.$$

$$(6) \quad \text{If } M = RA, |A| = n$$

let $N \subseteq M$ R -submodule.

$$\text{let } r + N \in M/N, \quad r = r_1a_1 + r_2a_2 + \dots + r_na_n + N$$

$$r = (r_1a_1 + N) + \dots + (r_na_n + N) = r_1(a_1 + N) + \dots + r_n(a_n + N)$$

$$M/N = R\{a_1 + N, \dots, a_n + N\}$$

$\forall a_i \in N$ then $a_i + N = 0 \Rightarrow M/N$ is generated by n or fewer elements.

$$\text{if } M = Ra \text{ then } M/N = R\{a + N\} \text{ or } R \cdot 0 \text{ (if } M = N)$$

$$(7) \quad \text{let } N \subseteq M \text{ } R\text{-submodule. suppose } M/N = R\{a_1 + N, \dots, a_n + N\} \text{ and } N = R\{b_1, \dots, b_k\}$$

$$\forall m \in M \text{ we have } m + N = \sum_{i=1}^n r_i(a_i + N)$$

$$\Rightarrow m = \sum_{i=1}^n r_i a_i + n \quad n \in N$$

$$\text{Since } N \text{ is finitely generated, } n = \sum_{j=1}^k r'_j b_j$$

$\Rightarrow M$ is finitely generated.

$$8. \quad S = \{(a_1, a_2, a_3, \dots) \in \mathbb{Z} \times \mathbb{Z} \times \dots \mid |a_i| < \infty\}. \quad [S = \mathbb{Z} \oplus \mathbb{Z} \oplus \dots]$$

$$\text{def. } +: S \times S \rightarrow S$$

$$(a_1, a_2, a_3, \dots) + (b_1, b_2, b_3, \dots) = (a_1 + b_1, a_2 + b_2, \dots)$$

$$\cdot: S \times S \rightarrow S$$

$$(a_1, a_2, a_3, \dots) \cdot (b_1, b_2, b_3, \dots) = (a_1 b_1, a_2 b_2, \dots)$$

S is a ring without identity since $(1, 1, 1, \dots) \notin S$

If S is finitely generated, then $\exists s_1, \dots, s_k$ s.t. $S = S\{s_1, \dots, s_k\}$.

$$\text{def. } \kappa: S \rightarrow \mathbb{Z}^+$$

$$\kappa(s) = \max\{i \mid s_i \neq 0\}$$

$$\forall s \in S, \kappa(rs_j) = \kappa(s_j) \quad \forall s_j \in \{s_1, \dots, s_k\}$$

$$\text{let } s_k \in \{s_1, \dots, s_k\} \text{ s.t. } \kappa(s_k) \geq \kappa(s_i) \quad \forall s_i \in \{s_1, \dots, s_k\}$$

However, $(0, 0, \dots, 1, 0, \dots, 0)$ where 1 is at $\kappa(s_k) + 1$ th position is in S which cannot be an $\mathbb{Z}$ -linear combination of s_i .

9. An R -module M is called irreducible if $M \neq 0$ and if $M, 0$ are the only submodules of M .

($\Rightarrow$) Suppose M is irreducible $\Rightarrow M \neq 0$ and $M, 0$ are the only submodules.

$$\text{let } a \neq 0 \in M$$

$$R[a] = M \text{ or } \{0\}. \text{ since } a \in R[a], R[a] \neq 0$$

$$\Rightarrow R[a] = M.$$

($\Leftarrow$) Suppose $M = R[a] \quad \forall a \neq 0 \in M$. Clearly, $M \neq 0$

Suppose $N \subseteq M$ is a submodule of $M, N \neq 0$

$$\text{let } n \in N, n \neq 0 \Rightarrow N = R[n] \subseteq N \Rightarrow N \text{ is irreducible.}$$

$$\mathbb{Z}/p\mathbb{Z} = \mathbb{Z}\{a\} \quad a \in \mathbb{Z}/p\mathbb{Z}, a \neq 0 \text{ are the only irreducible } \mathbb{Z}\text{-modules}$$

$$\text{Suppose } M \text{ is an irreducible } \mathbb{Z}\text{-module} \Rightarrow M = \mathbb{Z}\{a\} \quad \forall a \neq 0 \in M. \Rightarrow M \cong \mathbb{Z}/n\mathbb{Z} \text{ as } \mathbb{Z}\text{-modules.} \Rightarrow M \cong \mathbb{Z}/p\mathbb{Z}$$

10. Suppose R commutative with 1 . M be R -module.

Suppose M is irreducible. $\Leftrightarrow M = R[a]$ for any $a \neq 0 \in M$.

$$\text{let } \varphi_a: R \rightarrow M, \quad \varphi_a \in \text{Hom}_R(R, M) \text{ clearly.}$$

$$r \mapsto ra$$

$$\Leftrightarrow M \cong R/\ker \varphi_a$$

$$\text{Suppose } \ker \varphi_a \subseteq I.$$

$$\varphi_a(I) = I[a] \text{ is a submodule of } M \Rightarrow I[a] = 0 \text{ or } I[a] = M.$$

$$\Rightarrow I = \ker \varphi_a \text{ or } I = R$$

$$\Rightarrow \ker \varphi_a \text{ is a maximal ideal.}$$

(11) M_1, M_2 are irreducible R -modules,

Suppose $\varphi: M_1 \rightarrow M_2$ is a nonzero homomorphism.

$$\text{fix } a \neq 0 \in M_1$$

$$M_1 = R[a]$$

$$\varphi(a) = 0 \text{ then } \varphi(ra) = r\varphi(a) = 0 \text{ for all } r \in R. \Rightarrow \varphi = 0$$

$$\text{Suppose } \varphi(a) \neq 0 \Rightarrow M_2 = R[\varphi(a)]$$

$$\text{Suppose } m \in \ker \varphi, \quad \varphi(m) = \varphi(sa) = s\varphi(a) = 0 \Rightarrow s = 0 \Rightarrow m = 0$$

$$\text{Suppose } m' \in M_2, \quad m' = s\varphi(a) = \varphi(sa) \in \varphi(M_1) \Rightarrow \varphi \text{ is surjective}$$

$$M_1 \cong M_2.$$

$$\Rightarrow \text{If } M \text{ is irreducible, } \forall \varphi \in \text{End}_R(M) \Rightarrow \varphi^{-1} \in \text{End}_R(M) \Rightarrow \text{End}_R(M) \text{ is a division ring.}$$

(12) Let R be commutative. Let A, B and M be R -modules.

$$(a) \quad \text{Hom}_R(A \times B, M) \cong \text{Hom}_R(A, M) \times \text{Hom}_R(B, M)$$

$$\overline{\varphi}: \varphi \mapsto (\varphi \circ i_A, \varphi \circ i_B) \quad [\varphi(a, b) \mapsto (\varphi \circ i_A(a), \varphi \circ i_B(b))]$$

well defined map.

$$\text{let } \varphi, \psi \in \text{Hom}_R(A \times B, M) \text{ let } r \in R$$

$$\varphi + r\psi \mapsto ((\varphi + r\psi) \circ i_A, (\varphi + r\psi) \circ i_B)$$

$$\begin{aligned} \text{let } a \in A, \quad \varphi + r\psi \circ i_A(a) &= \varphi + r\psi(a, 0) = \varphi(a, 0) + r\psi(a, 0) \\ &= \varphi \circ i_A(a) + r\psi \circ i_A(a) \\ &= (\varphi \circ i_A + r(\psi \circ i_A))(a) \end{aligned}$$

$\Rightarrow \bar{\Phi}$ is an R -module hom.

Suppose $\varphi \in \ker \bar{\Phi}$

$$\bar{\Phi}(\varphi) = (\varphi \circ i_A, \varphi \circ i_B) = (0, 0)$$

$$\varphi|_{A \times \{0\}} = 0 \text{ \& } \varphi|_{\{0\} \times B} = 0 \Rightarrow \varphi = 0$$

$\Rightarrow$ Suppose $(\varphi, \psi) \in \text{Hom}_R(A, M) \times \text{Hom}_R(B, M)$

$$\text{consider } \varphi \circ \pi_A + \psi \circ \pi_B : A \times B \rightarrow M$$

$$\varphi \circ \pi_A : A \times B \rightarrow M \Rightarrow \text{well defined. } R\text{-module hom.}$$

$$\psi \circ \pi_B : A \times B \rightarrow M$$

$$\Rightarrow \varphi \circ \pi_A + \psi \circ \pi_B \in \text{Hom}_R(A \times B, M)$$

$$\begin{aligned} \bar{\Phi}(\varphi \circ \pi_A + \psi \circ \pi_B)(a, b) &= ((\varphi \circ \pi_A + \psi \circ \pi_B) \circ i_A(a), (\varphi \circ \pi_A + \psi \circ \pi_B) \circ i_B(b)) \\ &= (\varphi(a), \psi(b)) \end{aligned}$$

$\Rightarrow \bar{\Phi}$ is surjective $\Rightarrow \text{Hom}_R(A \times B, M) \cong \text{Hom}_R(A, M) \times \text{Hom}_R(B, M)$

(b) $\text{Hom}_R(M, A \times B) \cong \text{Hom}_R(M, A) \times \text{Hom}_R(M, B)$

$$\bar{\Phi} : \varphi \mapsto (\pi_A \circ \varphi, \pi_B \circ \varphi) \quad \varphi : M \xrightarrow{\varphi} A \times B \xrightarrow{\pi_A} A$$

$$\pi_A \circ \varphi \in \text{Hom}_R(M, A), \quad \pi_B \circ \varphi \in \text{Hom}_R(M, B)$$

Suppose $\varphi = \psi$ in $\text{Hom}_R(M, A \times B)$

$$\begin{aligned} \bar{\Phi}(\varphi)(m) &= (\pi_A \circ \varphi(m), \pi_B \circ \varphi(m)) \\ &= (\pi_A \circ \psi(m), \pi_B \circ \psi(m)) = \bar{\Phi}(\psi)(m) \Rightarrow \text{well defined.} \end{aligned}$$

$$\begin{aligned} \bar{\Phi}(\varphi + r\psi)(m) &= (\pi_A \circ (\varphi + r\psi)(m), \pi_B \circ (\varphi + r\psi)(m)) \\ &= (\pi_A(\varphi(m) + r\psi(m)), \pi_B(\varphi(m) + r\psi(m))) \\ &= (\pi_A\varphi + r\pi_A\psi, \pi_B\varphi + r\pi_B\psi) \\ &= ((\pi_A\varphi, \pi_B\varphi) + r(\pi_A\psi, \pi_B\psi))(m) \\ &= \bar{\Phi}(\varphi) + r\bar{\Phi}(\psi) \Rightarrow R\text{-module homomorphism.} \end{aligned}$$

Suppose $\varphi \in \ker \bar{\Phi}$ let $m \in M$.

$$\begin{aligned} \bar{\Phi}(\varphi)(m) &= ((\pi_A \circ \varphi)(m), (\pi_B \circ \varphi)(m)) = (0, 0) \\ \pi_B \circ \varphi(m) &= 0 \\ \pi_A \circ \varphi(m) &= 0 \quad \forall m \in M \Rightarrow \varphi = 0 \end{aligned}$$

Suppose $(\varphi, \psi) \in \text{Hom}_R(M, A) \times \text{Hom}_R(M, B)$

$$\varphi : M \rightarrow A \xrightarrow{i_A} A \times B$$

$$i_A \circ \varphi + i_B \circ \psi \in \text{Hom}_R(M, A \times B)$$

$$\begin{aligned} \bar{\Phi}(i_A \circ \varphi + i_B \circ \psi)(m) &= (\pi_A \circ (i_A \circ \varphi + i_B \circ \psi)(m), \pi_B \circ (i_A \circ \varphi + i_B \circ \psi)(m)) \\ &= (\pi_A \circ i_A \circ \varphi(m), \pi_B \circ i_B \circ \psi(m)) \\ &= (\varphi(m), \psi(m)) \Rightarrow \text{Hom}_R(M, A \times B) \cong \text{Hom}_R(M, A) \times \text{Hom}_R(M, B) \end{aligned}$$

13. let R be a commutative. let F be a free R module of finite rank.

$$\begin{aligned} \text{Hom}_R(F, R) &\cong \text{Hom}_R(R^n, R) \cong \text{Hom}_R(R, R) \times \dots \times \text{Hom}_R(R, R) \\ &\cong R \times \dots \times R = R^n \cong F \end{aligned}$$

$$R \rightarrow \text{Hom}_R(F, R)$$

$$\bar{\Phi} : r \mapsto rI$$

$$r + r' \mapsto r + r' \cdot I = (r \cdot I) + r' \cdot I$$

$$r \in \ker \bar{\Phi} \quad \bar{\Phi}(r) = rI = 0 \Rightarrow r = 0$$

Suppose $\varphi \in \text{Hom}_R(R, R)$

$$\text{if } \varphi(1) = 0 \text{ then } \varphi = 0, I$$

$$\varphi(1) = r \quad r \neq 0.$$

$$\forall r' \in R \quad \varphi(r') = r \cdot r' = r \cdot r' \cdot I = r \cdot I(r') \Rightarrow \bar{\Phi} \text{ is surjective } \Rightarrow R \cong \text{Hom}_R(R, R)$$

14. let R be commutative ring let F be free R -module of rank n .

$$\text{Hom}_R(F, M) \cong \text{Hom}_R(R^n, M) \cong \text{Hom}_R(R, M) \times \dots \times \text{Hom}_R(R, M) \text{ (n times)}$$

$$\text{Hom}_R(R, M) \cong M \Rightarrow \cong M \times \dots \times M \text{ (n times)}$$

(15). $e \in R$ is called idempotent (central) if $e^2 = e$, $e \in C(R)$

$$\text{let } m \in M \quad m = em + (1-e)m \Rightarrow M = eM + (1-e)M$$

$$x \in eM \cap (1-e)M$$

$$x = em = (1-e)m' \quad m, m' \in M$$

$$em - (1-e)m' = 0$$

$$\Rightarrow e(m+m') = m'$$

$$\Rightarrow e^2(m+m') = e(m+m') = em'$$

$$\Rightarrow e(m+m'-m') = em = x = 0.$$

$$\Rightarrow M = eM \oplus (1-e)M$$

(16) let $A_1, \dots, A_k$ be ideals in R . let M be an R -module.

$$\text{Def } \pi : M \rightarrow (M/A_1M) \times \dots \times (M/A_kM)$$

$$m \mapsto (m + A_1M, \dots, m + A_kM)$$

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$$0 \rightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \rightarrow \dots$$

$$\text{Tor}(\bigoplus_{i \in \mathbb{Z}^+} \mathbb{Z}/i\mathbb{Z}) = \bigoplus_{i \in \mathbb{Z}^+} \mathbb{Z}/i\mathbb{Z}$$

However, consider $z = (z_1, z_2, \dots) \in \prod_{i \in \mathbb{Z}^+} \mathbb{Z}/i\mathbb{Z}$, $\forall n \in \mathbb{Z}^+$ $n \cdot z = 0 \Rightarrow \text{Tor}(\prod_{i \in \mathbb{Z}^+} \mathbb{Z}/i\mathbb{Z}) \neq \prod_{i \in \mathbb{Z}^+} \mathbb{Z}/i\mathbb{Z}$.

Thus, $\bigoplus_{i \in \mathbb{Z}^+} N_i \neq \prod_{i \in \mathbb{Z}^+} N_i$.

21. Let I be nonempty index set. $\forall i \in I$ let N_i be a submodule of M_0 .

$$(i) \Rightarrow (ii) \text{ Suppose } \bigoplus_{i \in I} N_i \cong \bigoplus_{i \in I} N_i$$

Let $\{i_1, i_2, \dots, i_k\}$ be a finite set.

$$\text{Let } x \in N_{i_1} \cap (N_{i_2} + \dots + N_{i_k})$$

$$x = n_{i_1} = n_{i_2} + \dots + n_{i_k} \quad n_{i_j} \in N_{i_j} \quad \forall j \in \{2, \dots, k\}$$

$$-n_{i_1} + n_{i_2} + \dots + n_{i_k} = 0$$

$$-n_{i_1} + n_{i_2} + \dots + n_{i_k} \in \bigoplus_{i \in I} N_i \quad \exists (n)_I \in \bigoplus_{i \in I} N_i \text{ s.t. its all other components except } -n_{i_1}, \dots, n_{i_k} \text{ are zero.}$$

$$\Rightarrow \text{Since } \bigoplus_{i \in I} N_i \cong \bigoplus_{i \in I} N_i \Rightarrow -n_{i_1} = \dots = n_{i_k} = 0.$$

$$\Rightarrow x = 0$$

$$(ii) \Rightarrow (iii) \quad N_{i_1} + \dots + N_{i_k} \cong N_{i_1} \times \dots \times N_{i_k} \Rightarrow N_{i_1} + \dots + N_{i_k} = \bigoplus_{j=1}^k N_{i_j} \text{ (Prop 5)}$$

$$(iii) \Rightarrow (iv) \quad x \in \bigoplus_{i \in I} N_i$$

$x = \sum_{j \in J} n_j$ is a finite sum. $J \subset I$ finite. (by definition)

by (ii) each n_j is unique.

$$(iv) \Rightarrow (i) \quad \phi: \bigoplus_{i \in I} N_i \rightarrow \bigoplus_{i \in I} N_i \text{ is surjective } R\text{-module hom. (iii)}$$

$$\pi_i \circ \phi \rightarrow \sum_{i \in I} n_i$$

Suppose $\sum_{i \in I} n_i \in \ker \phi$

$$\sum_{i \in I} n_i = \sum_{j \in J} n_j = 0 \text{ by (ii) } n_i = 0 \quad \forall i \in I.$$

(22.) Let R be a P.I.D., let $M = \text{Tor}(M)$ be an R -module. Let p be a prime in R . $M_p = \{m \in M \mid \exists^+ n \text{ s.t. } nm = 0\}$

(a) $0 \in M_p$. Suppose $x, y \in M_p$ $r \in R$ $\exists^+ s, t > 0$ s.t. $r^s x = r^t y = 0$. Then $r^{s+t} p^{s+t} (x+y) = 0 \Rightarrow x+y \in M_p \Rightarrow M_p$ is a submodule.

(b) $\exists \text{ Ann}_R(M) = (0) \subseteq R$ (P.I.D.) $\Rightarrow a = u_1 p^{d_1} \dots p^{d_k} \quad p_i \text{ prime (u.f.d.)}$
 $M_p \subseteq M \Rightarrow (0) \subseteq \text{Ann}_R(M_p) \subseteq \text{Ann}_R(M) = (0) \Rightarrow p \nmid a \Rightarrow \text{f.v. } p_i \text{ for some } i \in \{1, \dots, k\}$
 $\forall e \in \mathbb{N}$

$$m \in M_{p_i} \quad \exists^+ d > 0 \quad p_i^d m = 0$$

$$\text{if } d < d_i \quad p_i^{d_i} m = 0 \Rightarrow m \in M_i$$

$$\text{if } d > d_i$$

$$d = d_i + r \quad d_i > r \quad p_i^d m = m_1 + \dots + m_k$$

$$p_i^d m = p_i^r (p_i^{d_i} m) = p_i^r (m_1 + \dots + m_k) = 0 \Rightarrow p_i^r m_1 = \dots = p_i^r m_k = 0 \Rightarrow m_1 + \dots + m_k = 0 \quad [M_i \cap M_j = \{0\} \text{ if } i \neq j]$$

$$\Rightarrow p_i^{d_i} m = 0$$

$$\Rightarrow m \in M_i \Rightarrow M_{p_i} = M_i.$$

(c) $\forall p \in R \quad M_p \subseteq M$ is a module. p prime

consider $I = \mathbb{Z}^+ \quad \{p_i\}_{i \in \mathbb{Z}^+}$ is a set of all primes in R . Let $M_{p_i} = M_i$

Suppose $\{i_1, \dots, i_k\} \subset I$ is a finite set.

$$\text{Let } x \in M_{i_1} \cap (M_{i_2} + \dots + M_{i_k})$$

$$x = m_{i_1} = m_{i_2} + \dots + m_{i_k}$$

$$\forall i_j \quad m_{i_j} \neq 0 \quad \exists^+ d_{i_j} > 0 \text{ s.t. } p_{i_j}^{d_{i_j}} m_{i_j} = 0$$

$$\text{then, } p_{i_1}^{d_{i_1}} \dots p_{i_k}^{d_{i_k}} x = 0$$

$$\Rightarrow (p_{i_1}^{d_{i_1}} \dots p_{i_k}^{d_{i_k}}, x) = 1.$$

$$\Rightarrow \exists r, s \in R \text{ s.t. } r p_{i_1}^{d_{i_1}} \dots p_{i_k}^{d_{i_k}} + s q = 1.$$

$$x = (r p_{i_1}^{d_{i_1}} \dots p_{i_k}^{d_{i_k}} + s q) x = r p_{i_1}^{d_{i_1}} \dots p_{i_k}^{d_{i_k}} m_{i_1} + s q (m_{i_2} + \dots + m_{i_k}) = 0$$

$$\Rightarrow M_{p_{i_1}} + \dots + M_{p_{i_k}} \subseteq M_{p_{i_1}} \oplus \dots \oplus M_{p_{i_k}}$$

$$\Rightarrow \bigoplus_{p \text{ prime}} M_p \subseteq \bigoplus_{p \text{ prime}} M_p$$

Suppose $m \in M$ $\exists r \in R$ s.t. $rm = 0$ $r = p_1^{d_1} \dots p_k^{d_k}$

$$\bigoplus_{p \text{ prime}} M_p = \{m \in M \mid rm = 0\} = M_1 \oplus \dots \oplus M_k \subseteq M_{p_1} \oplus \dots \oplus M_{p_k} \Rightarrow m \in \bigoplus_{p \text{ prime}} M_p$$

$$\Rightarrow M = \bigoplus_{p \text{ prime}} M_p$$

(23.) Let I be an index set. Let F_i be free modules.

$$\bigoplus_{i \in I} F_i \cong \bigoplus_{i \in I} F(A_i)$$

$x \in \bigoplus_{i \in I} F(A_i) \quad x = (f_i)_I = \sum_{i \in I} f_i \quad f_i \in F(A_i)$ is a finite sum. $\Rightarrow$ All but finitely many f_i are nonzero.

$x \in \bigoplus_{j \in J} F(A_j)$ $\exists$ decomposition in terms of the generating sets A_j and elements in R .

$$\Rightarrow \bigoplus_{i \in I} F_i \text{ is a free module.}$$

(24.) $\forall i \in \mathbb{Z}^+ \quad$ let N_i be the free $\mathbb{Z}$ -module $\mathbb{Z}$.

$$\text{Let } M = \prod_{i \in \mathbb{Z}^+} N_i.$$

each $m \in M$ can be uniquely written as $(a_1, a_2, \dots)$ with $a_i \in \mathbb{Z}$

$$\text{Let } N = \{x \in M \mid \text{finitely many nonzero}\}$$

Suppose M is a free-module with basis B

$$(n) \text{ let } S_i = \{(a_1, a_2, \dots, a_i, 0, 0, \dots) \mid a_i \in \mathbb{Z}\}$$

$$\text{claim: } N = \bigcup_{i \in \mathbb{Z}^+} S_i$$

$$\text{pf: } x \in N \Rightarrow x = (x_1, x_2, \dots) \text{ where } |\{i \mid x_i \neq 0\}| < \infty.$$

$$\text{let } k = \max\{i \mid x_i \neq 0\} \text{ which exists. } \Rightarrow x \in S_k \Rightarrow x \in \bigcup_{i \in \mathbb{Z}^+} S_i$$

$$\forall i \in \mathbb{Z}^+ S_i \subseteq N \Rightarrow \bigcup_{i \in \mathbb{Z}^+} S_i \subseteq N.$$

Each S_i is countable $\Rightarrow N$ is countable.

(b) N is countable, $N = \{x_1, x_2, \dots\}$ let $x_k \in N$

$$x_k = \sum_{b \in S_k} a_b \cdot b \quad a_b \in \mathbb{Z} \text{ by inclusion in } M. (S_k \subseteq B) \text{ since the sum is finite.}$$

$$\text{let } B_1 = \bigcup_{k \in \mathbb{Z}^+} S_k \text{ then } N \subseteq \mathbb{Z}B_1 = N_1$$

$$\text{claim: } N_1 = \bigcup_{k \in \mathbb{Z}^+} \mathbb{Z}S_k$$

$$\text{pf: } x \in N_1 \Rightarrow x = \sum_{i=1}^n a_i b_i \in \mathbb{Z}\{b_1, \dots, b_n\} \Rightarrow x \in \mathbb{Z}S_k \text{ for some } k.$$

$$\Rightarrow N_1 \subseteq \bigcup_{k \in \mathbb{Z}^+} \mathbb{Z}S_k$$

$\mathbb{Z}^+ \cong \mathbb{Z}S_k$ is countable $\Rightarrow N_1$ is countable.

(c) let $\bar{M} = M/N_1$, $\bar{M} = \{\sum_{i=1}^n a_i b_i \mid b_i \in B \setminus B_1, a_i \in \mathbb{Z}\} = \mathbb{Z}[B \setminus B_1] \Rightarrow \bar{M}$ is a free $\mathbb{Z}$ -module. [since M is free over B]

$$\exists \bar{x} \in \bar{M} \quad \bar{x} \neq 0 \quad \bar{x} = \sum_{i=1}^n a_i b_i + \bar{z} \quad \bar{z} \in N_1 \quad \bar{z} = \sum_{j=1}^l c_j s_j \quad s_j \in B_1$$

$$x = \sum_{i=1}^n a_i b_i + \sum_{j=1}^l c_j s_j$$

$$x_k = \text{gcd}(a_1, \dots, a_n, c_1, c_2, \dots, c_l) \mid x = k \cdot l$$

$$x = k \cdot l \left(\sum_{i=1}^n \frac{a_i}{k} b_i + \sum_{j=1}^l \frac{c_j}{k} s_j \right) + \sum_{j=1}^l c_j s_j$$

$$\bar{x} = k \cdot l \left(\sum_{i=1}^n \frac{a_i}{k} b_i \right) = k \cdot l \cdot \bar{m} \text{ for finitely many positive integers.}$$

$$\Rightarrow \bar{m} = \sum_{i=1}^n \frac{a_i}{k} b_i + \sum_{j=1}^l \frac{c_j}{k} s_j \in N_1 \text{ depends on } k.$$

And, clearly $\{k \cdot l\}_{l=1}^f$ is finite.

$$(d) \text{ let } S = \{(b_1, b_2, b_3, \dots) \mid b_i = \pm 1 \forall i\}$$

$$|S| = 2^{\aleph_0} > \aleph_0 \Rightarrow S \text{ is uncountable.}$$

N_1 is countable, $S \not\subseteq N_1 \Rightarrow \exists s \in S$ s.t. $s \notin N_1$

$$(e) \exists s \in S \quad s \notin N_1. \quad \forall k \in \mathbb{Z}^+ \quad s = (b_1, \dots, b_k, 0, \dots) + (0, \dots, b_{k+1}, \dots) = \bar{s} + q_k$$

$$s \in N \subseteq N_1 \Rightarrow \bar{s} = 0 \Rightarrow \bar{q}_k \neq 0.$$

$$\bar{q}_k = (0, \dots, b_{k+1}, b_{k+2}, \dots) = k(\alpha, \dots, \alpha_{k+1}, \alpha_{k+2}, \dots) = k \cdot \bar{m}_k \neq 0$$

$$\Rightarrow \bar{m}_k \neq 0$$

$$\Rightarrow \bar{s} = k \cdot \bar{m}_k \notin$$

(25) Suppose I is a nonempty index with partial order. For $i \in I$ let A_i be an R -module.

Further suppose, I is a directed set: $\forall i, j \in I \exists k \in I \quad i, j \leq k$

$$\forall i, j \text{ with } i \leq j \quad \exists p_{ij} \in \text{Hom}_R(A_i, A_j)$$

$$(i) \quad p_{jk} \circ p_{ij} = p_{ik}$$

$$(ii) \quad p_{ii} = \text{id}_{A_i} \quad \forall i \in I.$$

$$\text{let } B = \bigcup_{i \in I} A_i \quad (B, \sim) := a \sim b \Leftrightarrow \exists k \quad p_{ik}(a) = p_{ik}(b) \quad a \in A_i \quad b \in A_j$$

$$B/\sim = A = \bigcup_{i \in I} A_i$$

$$\text{Def } p_i: A_i \rightarrow A \quad p_i(a) = \bar{a}$$

$$R \supseteq A$$

$$r\bar{a} = \overline{ra}$$

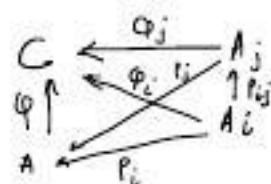
$$r(\bar{a} + \bar{b}) = \overline{r(p_{ik}(a) + p_{ik}(b))} = \overline{p_{ik}(ra) + p_{ik}(rb)} = \overline{p_{ik}(ra)} + \overline{p_{ik}(rb)} = r\bar{a} + r\bar{b}$$

$$(rs)(\bar{a}) = \overline{(rs)p_{ik}(a)} = \overline{p_{ik}(r(sa))} = \overline{r p_{ik}(sa)} = r \overline{p_{ik}(sa)} = r\bar{s}\bar{a}$$

$$(r+s)(\bar{a}) = \overline{(r+s)p_{ik}(a)} = \overline{r p_{ik}(a) + s p_{ik}(a)} = \overline{p_{ik}(ra) + p_{ik}(sa)} \\ = \overline{ra} + \overline{sa} = r\bar{a} + s\bar{a}$$

$\Rightarrow A$ is an R module. and $p_i \in \text{Hom}_R(A_i, A)$

Suppose C is any R -module. s.t. $\forall i \in I \exists q_i \in \text{Hom}_R(A_i, C)$ s.t. $q_i = q_j \circ p_{ij}$ whenever $i \leq j$



$$\text{Def } \varphi: A \rightarrow C$$

$x \in A_k \quad \varphi: \bar{x} \mapsto \varphi_k(x)$ is a well defined abelian group hom. (unique).

$$\varphi(r\bar{x}) = \varphi_k(rx) = r\varphi_k(x) \Rightarrow \varphi \in \text{Hom}_R(A, C). \text{ and } \varphi \circ p_i = q_i \quad \forall i.$$